The analog transformer: the human llm.

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Introduction:

What you are about to read is dramatization of a man with constrained resources some, very basic tools, an electric adder being the most advanced piece of equipment, and to a limited degree the help of an assistant who may do some small tasks that are straightforward but just too tedious to detail in frame. The professor sets the stage in an auditorium somewhere in the Midwest one dreary January day. He has arranged everything and simplified everything to a place where simplicity and functionality have come together in an armistice.

CHAPTER 1: THE ROOM AND THE REAM: SETTING THE STAGE AND INITIALIZING THE MACHINES

The oak desk hulks in the middle of the auditorium like a beached dreadnought, grain dark and scarred from decades of elbows and spilled coffee. Four sheets of bright white copy paper lie dead center, each already block-lettered in thick Sharpie whose solvent bite still prickles the sinuses:

DIMENSION ZERO
DIMENSION ONE
DIMENSION TWO
DIMENSION THREE

That's the cage we've built. d_model = 4. Four skinny numbers forced to shoulder queries, keys, values, attention heads, feed-forward layers, residuals—the whole transformer menagerie. No more dimensions; the desk can't take the weight. Anything wider and the paper starts to curl like it's trying to escape.

It's mid-January 2026, Midwest college auditorium, 1:37 p.m. CST by the wall clock whose second hand ticks like a nervous metronome. Outside the tall single window, yesterday's snow has melted into a steady grey drizzle that drums the glass in patient, endless code. Inside, a single 60-watt bare bulb swings on its cord overhead, throwing a jaundiced pool across the wood. Shadows stretch black from every tool, every stack. The air carries warm machine oil, the metallic tang of ribbon solvent, the faint must of old book paper, and the clean woody resin rising from the fishbowl of Go pieces to my right.

The bowl holds maybe two hundred black and white stones, river-smooth after centuries of tumbling in someone else's game. On top rests the hundred-sided die—translucent green plastic, numbers etched so faint you have to tilt it to catch the light. Randomness waiting its turn. Haven't rolled it yet. Haven't needed the chaos.

To the bowl's right squats the Sharp electric adding machine, faded yellow plastic, early-nineties vintage. Paper tape threaded and coiled in the cradle, ready to spit purple-inked verdicts. Switch is off; the motor silent for now. When it wakes it'll hum low and content, then whiiine sharp on equals, ka-ka-ka-ka we-we-we ka-ka as the tape advances and the ink blooms crisp. Tested it this morning: 1.234 + 5.678 = 6.912. Tape hissed out clean. Good enough for gradients later.

Beside it stands the IBM Selectric typewriter, buckling-spring legend. Keys gleam dull silver under the bulb. Each one a tiny coiled trap waiting to snap. Press a key: spring compresses, typebar arcs, slams through ribbon, thunk against paper, carriage ratchets left one space. Correction ribbon waits underneath—black top for typing, white bottom for lifting mistakes like they never happened. Whirr when you hit backspace, soft hiss, clean slate. The machine forgives without sermon.

I reach for the typewriter first. Slide a fresh sheet of bright white copy into the platen. Roller grips firm, no slip. Index three times—click-thunk, click-thunk, click-thunk—leaving breathing room at the top. Fingers hover. Then I type slow, deliberate, each letter a small violence: spring buckles under fingertip, typebar flies, ink slams through ribbon onto paper. Black blooms sharp.

Mistype the "R" in TRANSFORMER? Correction key whirrs—white ribbon rises halfway, presses down, lifts the error like breath on a mirror. Retype. Clean. No trace. The Selectric doesn't judge.

The primer waits at the desk's far edge. Fun with Dick and Jane, 1940s edition, cover soft and thumbed from a thousand small hands. I open it to the first spread. Black type on cream stock gone gentle sepia. Illustrations faded: Dick in shorts, Jane in a dress, Sally smaller, all wide-eyed at something ballooning upward. I read the text aloud once, voice low, letting the words settle into the room like dust:

See It Go.
“Look,” said Dick. “See it go. See it go up.”
Jane said, “Oh, look! See it go. See it go up.”
“Up, up,” said Sally. “Go, up, up, up.”
Simple. Repetitive. Engineered for children learning to map sound to symbol. Thirty-four word instances if you count every repeat. Twenty-nine unique if you collapse them. Our entire training corpus. No internet scrape, no Wikipedia dump, no Common Crawl. Just this. Thirty-four fragile sentences. Enough to break something open if we do it right.
The brass beam scale sits dead center now, pans level, needle pinned at zero. One black Go stone on the left pan, one white on the right—perfect balance. Later it'll weigh symbolic guilt in gradients, the slow drag of error pulling one side down. For now it's our conscience, level and cold.
I switch the adding machine on. Low hum rises, distant bees in a hive. Paper ribbon creeps forward with a soft whirr, purple ink cartridge primed. Ready.
I lean back. Chair wood creaks under me. Fingers smell of ribbon solvent and old ink. The drizzle outside has thickened to a steady hiss against the pane. The bulb buzzes faintly overhead, a tired insect trapped in glass.
This is war on the black box. Not with CUDA kernels or tensor cores. With paper, ink, thumbs, a dying adding machine, and whatever scraps of patience I've got left. We're going to feel every dot product, every gradient step, every update like physical labor—paper cuts, ink smudges, the ache in the wrist from too many manual multiplications.
The forward pass starts here. In this room. With these four white sheets marked DIMENSION ZERO through THREE. With this stupid little primer that taught a generation to read.
No GPUs. No backprop accelerators. Just friction. Just us.
I pick up the platinum desk pen—ultra-fine nib, the kind whose scratch travels up your spine like a wire. I underline the primer's first line once, slow:
See It Go.
Ink sinks in. Black on cream. Thought already shattering into shards.
The adding machine hums, patient. The typewriter waits. The scale balances.
Time to begin.

CHAPTER 2: THE CURRICULUM – RAW TEXT EXTRACTION AND NORMALIZATION

The primer lies open under the bare bulb like a confession I don't want to hear. Cream pages gone soft from decades of small fingers, black type still sharp enough to cut if you stare too long. No preface, no glossary, no mercy—just the first spread staring back. This is the curriculum. Our entire fucking universe compressed into thirty-four word echoes. Tiny enough to hold in one palm, stubborn enough to break something inside you if you try to force it into vectors by hand.
I don't rush. I can't afford to. The adding machine hums low in the corner, patient as a sleeping dog, tape coiled white and waiting. The Selectric sits silent, springs coiled, ready to snap. Drizzle hisses against the window like static on an old radio nobody's tuning anymore. I breathe in ribbon solvent, machine oil, old paper, the faint woody resin from the Go bowl. My fingers smell already of ink and regret.
First: extraction. Pure copy. No decisions yet. Just lift the text out of the book before the room decides to swallow it.

I slide fresh bright white copy into the platen. Roller grips with a soft crunch. Index twice—click-thunk, click-thunk—leaving space at the top like I'm afraid to crowd the words. Fingers hover, then fall:

RAW CORPUS EXTRACTION

FUN WITH DICK AND JANE (1940s edition)

“SEE IT GO” SPREAD – FULL TEXT AS PRINTED

Each key a small act of violence. Spring buckles under fingertip. Typebar arcs. Thunk. Ink slams through ribbon onto paper. Black blooms.

“Look,” said Dick. “See it go. See it go up.”

Jane said, “Oh, look! See it go. See it go up.”

“Up, up,” said Sally. “Go, up, up, up.”

Carriage ratchets left with every line. Mistype the exclamation after “Oh”? Whirr—correction ribbon rises slow, white bottom presses down like a gentle hand erasing sin. Retype. Clean. The Selectric forgives faster than I do.

I read it back aloud, voice gravel from too many late nights already, matching eye to finger, finger to voice:

“Look,” said Dick. “See it go. See it go up.”

Jane said, “Oh, look! See it go. See it go up.”

“Up, up,” said Sally. “Go, up, up, up.”

The title sits above like a header nobody asked for: See It Go. Three more tokens. Thirty-four total if you count every repetition without mercy. The primer doesn't lie. It repeats because children need the hammer. We need it too, apparently.

Now the knife: normalization.

Normalization is the process of converting raw, messy text into a consistent, simplified format by applying fixed rules—lowercasing all letters, stripping away punctuation, removing anything that isn't essential sequence—so the data becomes uniform and machine-digestible without irrelevant surface variations.

That normalization does the dirty work of stripping distractions the transformer has no use for—capitals that pretend at proper nouns, commas that fake grammar, exclamation marks that try to sell excitement—so attention can fall cleanly on word patterns, frequencies, and nothing else. No noise. Just signal, naked and cold.

I grab the red fountain pen—nib fine enough to feel every fiber of the paper resist—and rule the protocol across the top margin of a second sheet, letters deliberate so I can't pretend later I forgot:

NORMALIZATION RULES

* All lowercase. No exceptions.
* Strip all punctuation: quotes, periods, commas, exclamation marks—gone.
* No stemming or lemmatization. Keep the surface forms children actually see.
* Preserve every single repetition. Frequency is truth here.

Pencil first, testing. Slow scratches. Then platinum pen over the top, ultra-fine tip dragging like wire across bone. The normalized stream emerges, one word at a time, ink bleeding just a hair at the edges:

see it go look said dick see it go see it go up jane said oh look see it go see it go up up up said sally go up up up

I let the pen rest. Ink settles. The line stares up—thirty-four tokens, flat, relentless. No capitals to signal importance. No punctuation to break the chant. Just the raw loop of the primer, stripped to bone.

Now the normalization's done. We've got this clean, unadorned stream lying on the desk—thirty-four fragile units, no hiding places left. The repetition lands heavier now: “see it go” circles like a tired prayer. “up” stacks toward the end, insistent, almost pleading.
To feel the shape without crushing you under numbers, I speak only two counts into the quiet room while the pen hovers: “see” appears five times. “up” appears eight. Those two carry the weight of the whole thing. The rest press in behind them, silent witnesses.
The scale needle twitches as if it senses the small violence I just did to the text. I nudge the black Go stone a fraction closer to center. Needle settles. Balance restored—for now.
The adding machine waits, motor humming low, tape ready to judge whatever comes next. Drizzle drums steady. The bulb buzzes overhead, faint and tired like everything else in this room.
The corpus is born. Raw lifted. Normalized carved. Thirty-four tokens waiting on bright white paper, ink still faintly warm under my thumb.
Next: tokenization. Breaking the stream into shards small enough to number, to embed, to feed into the machine we've built from paper and patience.
I lean back. Chair creaks. Lungs full of solvent and dust. The forward pass feels farther away than ever.
But we keep going. Because stopping would be worse.
…….....
CHAPTER 3: TOKENIZATION – Chopping Words into Shards**

The normalized sheet lies pinned under the brass lamp on the oak desk, its single long line of lowercase words looking oddly exposed in the hard yellow light.

look said dick see it go see it go up jane said oh look see it go see it go up up up said sally go up up up

The ink is still slightly glossy in places where the typewriter ribbon was a touch too wet; tiny halos around a few letters. It's 1:53 p.m. The rain outside has settled into a low, steady drumming—tap-tap-tap-tap-tap—like someone impatiently waiting for an answer on the other side of the window. Inside, the room has grown noticeably warmer. The adding machine's standby hum is a constant low throb now, almost like a heartbeat. The typewriter sits a few inches away, its metal frame still radiating faint heat from the last typing pass. My fingers smell of ribbon solvent and the iron tang of the black fountain pen ink.

Tokenization is the next unavoidable step. Before we can assign numbers, before we can build any matrices or run a single dot product, we have to break this continuous stream into discrete, indivisible units the model can actually process. And because this is a new concept, we stop here and give it the full thrice treatment—no skipping, no summary, no mercy.

First—the crisp, dictionary definition: Tokenization is the process of segmenting a continuous stream of text into smaller, meaningful units called tokens, which serve as the fundamental, atomic elements for all further linguistic and mathematical processing in a transformer model.

Second—the functional warmth: Tokenization is what allows unstructured human language to be transformed into a finite sequence of discrete items that can be mapped to integers, embedded into dense vectors, and subjected to linear algebra operations. Without clean token boundaries, the entire forward and backward pass collapses into chaos; the adding machine and the paper matrices would have nothing solid to grip, no way to compute attention scores or gradients.

Third—the casual familiarity: Tokenization is just us taking a sharp pair of scissors and carefully cutting the sentence into a neat pile of word-bricks so the Sharp can count them one by one without choking on the glue.

For our tiny toy universe today, the decision is straightforward: word-level tokenization. No subword merges, no byte-pair encoding, no character splitting. The primer is written in short, whole words with almost no morphology—nothing begs to be broken further. If we went character-level we'd turn twenty-nine tokens into over a hundred individual letters, and our d_model=4 context window would suffocate before we even started. Word-level keeps the sequence length manageable and the math honest.

I reach for the black fountain pen again—the fine steel nib that makes a soft, rasping scratch against the cheap copy paper. (The nib glides smoother on the primer's cream stock, but this ledger paper is rougher, more absorbent; the ink bleeds just a hair if I press too hard.) I begin at the left end of the normalized line, underlining each word the first time I see it. Repeats get skipped. My hand wants to underline every single "see" by instinct—years of writing linear sentences have trained the muscle—but logic overrides. The pen hovers. A tiny bead of black ink pools at the tip, threatening to drop and ruin the line. I pull back just in time.

look — clean horizontal stroke, nib scratching faintly.
said — underline.
dick — underline.
see — underline.
it — underline.
go — underline.
see — my hand twitches, almost underlines again. I force it to hover. The ink bead trembles, then recedes. Skip.
it — skip.
go — skip.
up — new. Underline.
jane — underline.
said — skip.
oh — underline.
look — skip.
see — skip.
it — skip.

go — skip.
see — skip.
it — skip.
go — skip.
up — skip.
up — skip.
up — skip.
said — skip.
sally — underline.
go — skip.
up — skip.
up — skip.
up — skip.

One deliberate pass. Ten unique words underlined:

look said dick see it go up jane oh sally

That's the vocabulary seed. Ten distinct token types. The full normalized sequence still contains twenty-nine token instances because of repetition—"up" appears seven times (Jane's "up" once, Sally's "up, up" twice, Sally's "up, up, up" three times—total seven), "see" four times, "it" four times, "go" four times, "said" three times. Those frequencies are preserved in the original line; we don't erase them. We just collect the dictionary once.

I lean forward, elbows on the desk, and read the underlined words aloud in order, voice low and gravelly from the dry room air:

"look... said... dick... see... it... go... up... jane... oh... sally."

It sounds almost like a roll call in an empty schoolhouse. Each word now tagged as unique. The repetitions—Sally's insistent "up up up"—stay alive in the sequence because that's where the pattern lives. Collapse them and the model never learns escalation, never feels the childish urgency. Tokenization doesn't destroy frequency; it exposes it for later counting.

I thread a fresh bright white sheet into the IBM Typewriter. Header, all caps, typed slowly:

TOKENIZATION – WORD-LEVEL UNIQUE TOKENS
(EXTRACTED FROM NORMALIZED CORPUS – 10 DISTINCT TYPES)

Then type them one per line:

look
said
dick
see
it
go
up
jane
oh
sally

Each entry: spring buckles, typebar arcs, ink blooms, carriage advances with a soft ratchet. The "y" in "sally" feels slightly sticky—maybe needs oil—but it thunks home clean. When the last line is done I tear the sheet free with a rip that echoes in the quiet room and lay it beside the growing stack.

Five layers on the desk now:

1. Open primer – polite original with punctuation and capitals.
2. Raw extraction – typed preserving formatting.
3. Normalized corpus – lowercase, punctuation stripped.
4. Unique tokens underlined on normalized sheet.
5. Typed unique token list – ten clean entries.

The electric adding machine hums on, patient. The beam-and-spring scale needle pinned at zero, nickel and dime level. The Go bowl sits untouched, black stones glossy, white stones soft in the light. The hundred-sided die rests on top like a quiet promise.

Outside, the rain has eased to occasional drips, plinking against the sill. Inside, the bulb buzzes, the typewriter cools, my fingers ache faintly from pen and keys. Shoulders tight from the slow, focused posture.

We've chopped the text into shards. Ten unique pieces collected. Twenty-nine instances preserved. The primer's rhythm is now discrete, countable, ready for IDs.

The black box opens another careful inch. One honest, underlined stroke at a time.
……
CHAPTER 4: VOCABULARY CONSTRUCTION – Building the Unique Word List and IDs

The ten unique tokens sit typed on their fresh white sheet, one per line, lined up under the brass lamp like prisoners in a holding cell:

look
said
dick
see
it
go
up
jane
oh
sally

It's 2:01 p.m. sharp. The rain has finally quit; only the occasional drip plinks off the gutter outside the window now, a slow, irregular metronome. Inside, the bulb buzzes its steady complaint, the yellow light catching motes of dust that drift lazily above the typewriter. The adding machine's low hum has become background noise—almost soothing after an hour of it. The room smells of cooling ribbon, fresh black ink, the faint iron bite of fountain-pen ink on my fingers, and the clear Go bowl that still sits untouched to my right. The hundred-sided die perches on top of the stones like a tiny green cannonball, waiting for its cue.

Vocabulary construction is the bridge we have to cross now. We've collected the unique pieces; next we tag each one with a permanent integer ID starting from zero. And because this is a new concept—assigning stable integer IDs to tokens—we stop right here and give it the full thrice treatment.

First—the crisp, dictionary definition: Vocabulary construction is the process of creating a fixed, closed set of unique tokens from the corpus and assigning each one a unique integer identifier (typically starting at 0) that serves as its permanent numerical representation throughout all model operations.

Second—the functional warmth: The vocabulary with its integer IDs turns words into something the mathematics can manipulate—vectors, matrices, dot products, gradients. Without these fixed numbers, there's no consistent lookup table for embeddings, no way to one-hot encode, no reliable way to compute similarity or loss. It's the moment language stops being poetry and becomes arithmetic fuel for the adding machine and the paper stacks.

Third—the casual familiarity: Vocabulary construction is just us giving each word its own dog tag—0 through 9—so when we later shove them through projections or attention, the Sharp knows exactly which number to punch and the matrices know exactly which row to multiply.

I slide the unique-token sheet back under the typewriter platen—no new paper yet; I want the IDs right beside the words for instant reference. Header extension, typed slow and deliberate:

VOCABULARY CONSTRUCTION – ASSIGNING 0-BASED INTEGER IDs
(ORDER OF FIRST APPEARANCE – DETERMINISTIC, NO SORTING)

Then I type the ID before each word, using tab for clean columns:

0 look
1 said
2 dick
3 see
4 it
5 go
6 up
7 jane
8 oh
9 sally

Each number strike feels heavier than the letters: shift for 0, tab, word. The buckling springs snap back with that familiar click-thunk. The "2" on "dick" lands a hair crooked—ribbon a bit dry—but I let it stand. Imperfections are honest. No correction ribbon this time; we live with the small flaws.

Why first-appearance order? Because it's simple, repeatable, human-doable without extra passes. Alphabetical would require sorting the list by hand—tedious and error-prone on ten items. Frequency ranking would demand counting first (we'll do that next). First-seen is what my eyes and memory naturally provide as I read the normalized line. "look" hits first, so 0. "said" next, 1. "dick" third, 2. And so on until "sally" claims the last slot at 9.

I lean back, chair creaking, and stare at the list. Arbitrary, yes. But in the vector space these numbers will soon inhabit, 9 is literally farther from 0 than 2 is. Does that mean the model will initially treat "sally" as more "distant" from "look" than "dick" is? Probably. Until gradients nudge the embeddings around. The thought makes me smile faintly—our tiny universe already carries the seeds of bias just from the order we met the words. Computational linguistics in microcosm.

I pull the updated sheet free with a rip that echoes louder than usual in the quieted rain. Lay it beside the stack. Six layers now:

1. Primer original – polite, punctuated.

2. Raw extraction – typed original formatting.
3. Normalized corpus – lowercase, stripped.
4. Unique tokens underlined on normalized sheet.
5. Typed unique token list.
6. Vocabulary with 0–9 IDs assigned.

Next, frequency tally—to feel the weight of repetition before we vectorize.

Fresh white sheet threaded into the Typewriter. Header:

TOKEN FREQUENCY COUNT – FROM NORMALIZED CORPUS

Then list each vocab word and its count, glancing back at the normalized line:

look – 2
said – 3
dick – 1
see – 4
it – 4
go – 4
up – 7
jane – 1
oh – 1
sally – 1

Typing each tally: word, tab, dash, number. "up" gets the longest mark—seven instances (Jane's single "up", Sally's "up up", Sally's "up up up"—exact match to the source). The carriage ratchets left with every entry. "up" dominates the signal; Sally's enthusiasm is the loudest echo in this tiny corpus.

Sheet pulled free. Laid on top. The repetitions are inked now, undeniable. The model will learn far more from those sevens and fours than from the lonely ones.

The adding machine hums on, still quiet—no arithmetic required yet. The beam-and-spring scale needle rests at zero, nickel and dime balanced. The Go bowl sits patient, stones cool and smooth. The hundred-sided die waits.

Outside, the drizzle has stopped completely; faint light from a breaking cloud filters through the window. Inside, the bulb buzzes, ink lingers on my fingers, shoulders ache from the slow hunch.

We've built the vocabulary. Ten words, ten IDs, frequencies tallied. Language numbered, tagged, ready for one-hot vectors.

The black box cracks wider. One numbered, honest step at a time.
……

CHAPTER 5: ONE-HOT ENCODING – Flipping Switches for Sparse Vectors

Two seventeen p.m., January fourteenth, twenty-six. The room feels smaller somehow. The bare bulb's yellow pool tightens around the oak desk like it's trying to hold warmth. Rain's long gone. Outside the window only flat grey sky remains, and the slow drip-drip from the gutter. Inside, the Sharp adding machine hums its low contented drone. Haven't touched it yet today. Paper tape lies coiled patient. The IBM sits silent for the moment. Carriage returned. Fresh ribbon smells faintly of solvent. The fishbowl of Go pieces rests cool to my right. Black and white stones smooth under idle fingers I haven't quite let wander there. The hundred-sided die still perches on top. Green plastic catches the light like a promise of chaos later.

We've got our vocabulary nailed down. Ten tokens. Zero through nine. Frequencies tallied in black ink. Next job: turn each word into something the mathematics can actually chew on. Not embeddings yet. That's the step after. First we need one-hot encoding. New term. New concept. So we stop dead and give it the full thrice treatment. Right here. Right now.

First—the crisp dictionary definition: One-hot encoding is a technique that represents a categorical variable such as a token ID as a sparse binary vector. Length equals the vocabulary size. All elements stay zero except for a single one at the index that matches the token's ID.

Second—the functional warmth: In our tiny transformer, one-hot vectors act as perfect, unambiguous switches. Each word lights up exactly one position in a long row of lights. The rest stay dark. This sparsity lets us do clean, exact lookups in the embedding table later. No fuzzy overlap at the start. It's the cleanest way to tell the adding machine and the paper matrices "this is look, nothing else, full stop."

Third—the casual familiarity: One-hot encoding is just us flipping a single switch for each word. Ten possible switches in our vocab. Only one gets thrown per token. Simple. Brutal. Honest. No ambiguity. No half-measures. The Sharp will love it when we start multiplying.

I reach for a fresh sheet of bright white copy paper. Seventh layer now building on the desk. Thread it into the typewriter. Platen grips firm. Index twice—click-thunk, click-thunk—then type the header slow and deliberate:

ONE-HOT ENCODING – EXAMPLE VECTORS FOR VOCABULARY SIZE TEN

Beneath it, I type the skeleton first. Leave space for the actual vectors:

Token	ID	One-Hot Vector
look	zero	[one followed by nine zeros]
said	one	[zero one followed by eight zeros]

dick two [two zeros one followed by seven zeros]
see three [three zeros one followed by six zeros]
it four [four zeros one followed by five zeros]
go five [five zeros one followed by four zeros]
up six [six zeros one followed by three zeros]
jane seven [seven zeros one followed by two zeros]
oh eight [eight zeros one followed by one zero]
sally nine [nine zeros one]

Each bracketed vector gets typed one digit at a time. Buckling springs snap with every key—click-thunk for the ones, click-thunk for the zeros. Carriage ratchets left steady. Mistype a zero on "up"? Whirr of the correction ribbon. White bottom lifts the error clean away like it never existed. Retype. Perfect.

In the mind's eye these aren't just numbers on paper. They're rows of ten tiny brass switches mounted on a long wooden board. Each switch wired to a different bell. When "look" comes along, only the zeroth switch flips up with a metallic snick. The rest stay down. When "up" arrives—Sally's favorite—the sixth switch snaps upright while the others sleep. Ten switches. Ten bells. Only one rings per word. Clean. Absolute.

I pull the sheet free with a soft rip. Lay it beside the vocabulary list. The rainbow ream waits in the corner—red for Queries, blue for Keys, green for Values, yellow for Feed-Forward, violet for gradients—but we haven't earned those colors yet. White copy still rules the day.

To feel the sparsity in my bones I decide to write out one full example from the normalized corpus by hand with the fountain pen. Black ink. Underlining the single one in each vector so it stands out like a lit match in the dark. The pen nib scratches faint against the page. Ink smells sharp and metallic. My thumb already has a small black smudge from steadying the paper.

This sparsity is brutal efficiency. Most positions dark. Only one lit per token. When we slide these vectors into the embedding lookup table next chapter, most of the multiplications will be zero. Free work. The adding machine won't even break a sweat on those. But the single lit position? That one carries the whole identity of the word forward into four-dimensional space. One number. Multiplied four times—once per dimension—becomes the start of everything.

I lean back. Chair creaks. Shoulders ache from the hunch. The bulb buzzes overhead like it's tired too. The Go bowl sits untouched. Stones cool, waiting. The hundred-sided die gleams patient. The beam-and-spring scale needle rests dead level—nickel and dime still balanced, no gradients to weigh yet.

Outside, the grey sky hasn't budged. Inside, the room smells of ink, ribbon solvent, warm machine oil, and the glint from the fishbowl. We've turned words into switches. Language flipped into binary torches—one flame per token, the rest dark. The black box cracks another sliver.

Next we'll give those torches actual light—embeddings, random floats pulled from thin air and pinned to paper. But that's for the next sheet. The next hour.

For now I just sit with the one-hot vectors staring back at me. Ten lonely ones on ten long rows of zeros. Simple. Honest. Ready.

There we go—no sentence breaks the cap, numbers spread thin and deliberate across lines and clauses. All the tactile meat preserved, rule obeyed in letter and spirit. If any other chapter's still got a sneaky cluster hiding, point it out and I'll scrub it clean same way. Ledger stays straight.

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CHAPTER 6: EMBEDDING TABLE INITIALIZATION – Random Floats and the Lookup Matrix

It is late afternoon. The rain has stopped. The silence left behind is heavy. It feels like a wet wool blanket draped over the auditorium.

On the oak desk, the stack of One-Hot vectors sits perfectly neat. Ten sheets. Ten rows of zeros. And ten lonely ones.

They are precise. They are brittle. And they are utterly useless for poetry.

A One-Hot vector is a switch. It says "I am Look." It says "I am not Dick." That’s it. It has no opinion on whether "Look" is a verb or a noun. It doesn't know that "See" is a cousin to "Look." In this format, they are strangers. They stand in a dark room exactly 1 meter apart.

To make them speak, we need to give them a body. We need to turn that single, hard switch into a rich, dense ghost.

We need the Embedding Table.

I shove the Sharp adding machine aside. It’s no use here. This isn’t arithmetic yet; it’s genesis. I reach for the fishbowl. The Go stones clatter, cool and mineral. I dig past them.

I pull out the hundred-sided die.

It is green translucent plastic. It is the size of a golf ball. It is practically a sphere.

I hold it carefully. You don't "roll" a Zocchihedron. Not unless you want to chase it under the radiator. It hates friction. It wants to roll forever. If you throw it, it runs away. You have to corral it.

I take the lid of a shoebox from the floor. I place it on the desk. This will be the pen.

This sphere is our chaos engine. Before the model learns anything, it must start with random noise. We have to hallucinate a personality for these words before we can teach them to be right.

New concept. Stop. The Thrice Rule applies.

First—the crisp, dictionary definition: An Embedding Table is a learnable lookup matrix. Its size is the Vocabulary count by the Model dimension. It initializes with small random continuous values. It maps each discrete integer token ID to a dense vector. This allows the model to capture relationships in a continuous space.

Second—the functional warmth: Think of the One-Hot vector as a coat check ticket. It is just a piece of paper with the number 1 on it. You can't wear the ticket. The Embedding Table is the back room where the coats hang. You hand over the ticket. The clerk hands you back a heavy, wool trench coat. The coat has texture. It has weight and warmth. The ticket was just the claim check; the embedding is the object itself.

Third—the casual familiarity: We are trading a stick for a lump of clay. The One-Hot vector is a rigid stick pointing north. You can't mold it. The Embedding vector is a handful of wet clay. Right now, the clay is just a shapeless blob we pulled out of the riverbank. But because it's clay, we can eventually squish it. We can twist it until it looks like a word.

I pull a fresh sheet of bright white copy paper. It is layer 8 on the desk now. I thread it into the IBM Selectric.

Header, deliberate:

EMBEDDING TABLE. RANDOM INITIALIZATION.
10 Words by 4 Dimensions.

I'm not going to type the numbers. I'm going to draw the grid. The fountain pen scratches 4 vertical lines down the page. 4 columns. 10 rows.

40 empty cells. 40 empty souls waiting for a number.

I look at the first row. Token 0 is "Look."

I need 4 numbers. 4 distinct personality traits. We want small numbers centered around zero.

I place the green sphere in the shoebox lid. I nudge it gently.

It swirls around the rim. It refuses to stop. It creates a low, plastic hum against the cardboard. Finally, it settles.

Call it 0.072. Positive.

I write it in the first box. This is Dimension 0. In my mind's eye, I'm building a sculpture out of junk. I look at the junk loom I've rigged up. Dimension 0 is the **Shoelace**.

So, for the word "Look," the shoelace is pulled a little tight. Not strangling, just snug.

I nudge the sphere again. Gently.

18. But let's make this one negative. Write -0.018.

Dimension 1 is the **Piano Wire**. For "Look," the piano wire is slack. It is barely humming. A negative value means it is dipping below the horizon.

Nudge.

91. High. Let’s keep it positive. Write 0.091.

Dimension 2 is the **Spaghetti**. The noodle is arched high. It is brittle. "Look" has a high value here.

Nudge.

44. Negative. Write -0.044.

Dimension 3 is the **Phone Cable**. It is curled back on itself.

So the vector for "Look" is set. Four numbers.

It means nothing. It is pure noise. But it is *unique* noise. No other word will start with exactly this shape. "Look" now has a fingerprint.

I lean back. My wrist pops. Doing this 40 times is madness. If I drop the die, it’s gone.

I look at the corner. The Assistant.

She is watching the sphere. Her eyes are like apertures. They dilate and contract with the light. She doesn’t breathe. I haven't seen her chest move in 1 hour.

I slide the paper and the shoebox lid across the oak desk toward her.

"Gaussian distribution," I say. "Mean zero. Variance 0.02. Fill the table."

She reaches into the box. She picks up the sphere. She doesn't roll it. She cups it in her hands. She shakes it. It sounds like dried teeth rattling in a jar. She shakes it so fast the sound becomes a continuous hiss.

She sets it down gently. She writes a number. Shakes. Sets. Writes.

She treats it like a bubble that might burst. She is moving faster than the Selectric could type.

In 5 minutes, she slides the sheet back.

The grid is full. 40 numbers. A mix of positives and negatives. They are mostly small, jagged, and random.

Row 1 is "Said." It has 4 random values.
Row 6 is "Up." It has 4 other random values.

I stare at the row for "Up." That vector is going to cause us so much trouble later. Right now, it is just random noise. But eventually, the gradients are going to grab that vector. They will yank it toward the sky.

But how do we use this table?

This is the final piece of the mental sculpture. We have the One-Hot vector from Chapter 5.

Mathematically, we perform a matrix multiplication. We take that row of numbers and crash it into this new table.

But look closer.

If you multiply a vector that is all zeros and a single 1 against a matrix, you aren't really doing math. You are doing a heist.

The 1 in the first position acts like a flashlight beam. It shines on the first row of the Embedding Table. The zeros keep the other 9 rows in the dark.

We don't add. We don't multiply. We just *copy*.

The One-Hot vector is a selector switch. It says "Give me Row 0."

It is a lookup.

I place the One-Hot sheet from Chapter 5 next to this new random grid.
On the left: The Switch.
On the right: The Payload.

When the machinery starts, the first token "Look" slides into the slot. It will trigger the switch. The switch will grab Row 0 from this table. And that ghost is what will float down the river into the Transformer.

It is not a word anymore. It is a coordinate in 4 dimensions.

Right now, since the numbers are random, the word "Look" is floating somewhere in the fog. It is totally lost. "Up" is floating nearby. It is equally lost. They don't know they are related. They don't know anything.

They are just raw coordinates waiting for a map.

I pick up the sheet. The ink is still wet. The numbers look like tiny insects trapped in the grid.

The Assistant has returned to her stillness. The sphere sits in the box lid. It is green and silent.

We have our embeddings. The words have bodies.

But bodies are useless without time. "Look" means nothing if we don't know *when* it was said.

I look at the heavy book of mathematical tables on the corner of the desk. The sine waves are waiting in there.

"Time to give them a wristwatch," I whisper.

I reach for the next sheet. **Positional Encoding.**

The black box cracks a little wider.
…….
CHAPTER 7: POSITIONAL ENCODING – Injecting the Arrow of Time

It is 3:12 p.m., January 14, 2026. The bare bulb's yellow has thickened into something almost syrupy. The shadows from the oak desk stretch longer across the floorboards.

Outside the single window, the Nebraska grey holds steady. No wind. No break. Just that flat, pressing quiet.

The Sharp adding machine hums low in standby. The paper tape is coiled and patient. The Typewriter carriage sits returned, the smell of ribbon solvent fading but still sharp under the warm oil smell.

The fishbowl of Go pieces rests cool to my right. The black and white stones are smooth. The hundred-sided die is perched quiet after its earlier work on the embeddings.

We've given our ten tokens their first bodies. We have four-dimensional vectors. We have random floats pinned to paper in ten neat rows.

But here is the rub: transformers have no native sense of order.

"Look said Dick" reads the same to the model as "Dick said Look". Unless we force the sequence into the math.

Words arrive as a bag. We need to stitch time back in. Enter **Positional Encoding**.

New term. New concept. Stop dead. Give it the full Thrice Rule on the oak.

First—the crisp dictionary definition: Positional encoding is a fixed set of deterministic vectors added to the token embeddings to inject information about the absolute or relative position of each token in the input sequence. In the original transformer, these are sine and cosine functions of different frequencies applied to each dimension.

Second—the functional warmth: Without positional encoding the model treats every position the same. Order vanishes. Sentences collapse into word soup. These encodings act like timestamps stamped onto each token's vector. They ensure "see it go" stays meaningfully different from "go it see".

Third—the casual familiarity: Positional encoding is just us gluing a little "I'm the third word" badge onto each embedding vector. It is fixed math. No learning is needed at the start. Sine and cosine do the heavy lifting so the adding machine can add them straight in without drama.

Since our sequences are short, we could hard-code positions 0 through 9. But fidelity matters. We will do it the classic way: sine and cosine functions across the four dimensions.

I reach for a fresh sheet of bright white copy paper. It is the ninth layer now cluttering the desk. I thread it into the Typewriter.

Index three times. Click-thunk. Click-thunk. Click-thunk.

Header slow and deliberate:

POSITIONAL ENCODING
. Sequence positions: 0 to 9.

I need a 10 by 4 matrix. One row per possible position.

Forty numbers again. But deterministic this time. The Sharp adding machine is useless here; it knows only arithmetic, not geometry.

I shove it aside. I reach for the heavy brick of a book sitting on the corner: *Standard Mathematical Tables, 11th Edition*. The spine cracks as I force it open to "Natural Trigonometric Functions".

The pages smell of dry vanilla and basement dust.

I run my finger down the columns. Pure lookup work.

For position 0: The math is trivial. Sine of 0 is 0. Cosine of 0 is 1.
Row 0: `[0, 1, 0, 1]`.

I type it clean. Click-thunk for each digit.

Position 1.
Dimension 0 uses Sine. The denominator is 1. The argument is 1 radian. I flip pages. 57 degrees. 17 minutes.
Sine is 0.8415.

Dimension 1 uses Cosine. Cosine of 1 radian is 0.5403.

Dimension 2. The frequency changes. Denominator is 100. The argument is 0.01 radians. Sine is roughly 0.0100.

Dimension 3. Cosine is roughly 1.0000.

I find the values in the grid of tiny numbers. My finger traces the line so I don't lose the place.

0.8415. 0.5403.

Then the other two values,
0.0100. 1.0000.

I type them into the matrix row. Click-thunk. Click-thunk.

Position by position. Row by row.

Position 2. Sine of 2. Cosine of 2. I have to interpolate between degrees in the book. I scribble calculations in the margin with the fountain pen.

It takes thirty-five minutes of page-flipping and typing. Fingers cramp. Shoulders knot tighter.

Mistype a 0.4161 as 0.4610? Whirr of correction ribbon on the Typewriter. The white bottom lifts the error like breath on glass.

Retype. Clean.

In the mind's eye these rows aren't numbers. They are gentle waves rippling along four strings. Higher frequencies twitch faster in the later dimensions.

Position 0 sits flat and calm. Position 9 oscillates a touch more wildly.

When we add them to the embeddings next chapter, each token's vector gets tilted by its place in line. It is like pinning a tiny compass needle to every word. The model knows which way time flows.

I pull the sheet free with a soft rip. I lay it beside the embedding table.

The two matrices stare at each other. Random floats on one. Deterministic waves on the other.

Soon they'll sum. Token by token.

The chair creaks as I lean back. The Go bowl is untouched. Stones cool. Die quiet.

The room smells of warm oil. Fresh purple tape ink. Ribbon solvent. Metallic fountain-pen tang. And the dust of the math tables.

Outside, grey presses on. Inside, we've stitched time into the math. Order returns.

The black box cracks another sliver.

Next we'll assemble the full input matrix. Embeddings plus positions. Ready for the rainbow projections.

For now I sit with the positional waves rippling on paper. Forty numbers. A heartbeat for sequence.

Fragile. Precise. Ours.

…….

CHAPTER 8: INPUT MATRIX ASSEMBLY – Adding Meaning to Time

“Three forty-five p.m. dead on, January fourteenth, twenty-six. Rain's buggered off, left this thick damp quiet leaning on the window like a tired dog. Room's cooled right through now, bulb dangling slow, spilling that jaundiced yellow over the oak scars. Eyes gritty as hell; forearms throbbing where the desk's bitten in too long. Adding machine's silent, yellow plastic still holding faint warmth. IBM keys cold and waiting, carriage primed. Brass scale level as conscience—one black stone, one white—needle pinned zero. Fishbowl stones smell clean resin, die on top like chaos taking a nap.

Two stacks facing off across the wood. Left: embeddings from Chapter Six—random floats in four thin dimensions, the rough soul of each word from our thirty-four fragile lines. Right: positional sines and cosines from Seven—steady waves marking spot in line. Separate they're dead in the water. Word without place is scattered noise; place without word is an empty bench. To fire up the forward pass, we fuse them. Element-wise sum. No stretching vectors, no clever hacks—just addition. Input Matrix. New player. Thrice rule.

First, crisp: add Token Embedding Matrix to Positional Encoding Matrix, piece by piece. One four-float vector per token. Each row packs word meaning and sequence position tight.

Second, warmth: stack the reams, hold 'em to the bulb. Light bleeds through. Sine waves on top rock the random floats underneath—gentle shove here, small tilt there. Enough to murmur: this ain't abstract 'Look'; it's 'Look' rooted firm at the head of the queue.

Third, casual: handshake. Meaning meets moment. Numbers from left pile plus numbers from right, jot the result. Basic sum, honest friction.

Building it in your head—macaroni sculpture again. You're blind, hands gluey, pasta everywhere. Deaf woman opposite, gesturing broad and urgent, no fine print. She waves 'curve this way,' you stick a piece crooked, feel for balance, adjust when she signs again. Clumsy, but the form grows.

Take 'said' at position one. Embedding base feels random, wobbly curves. First bit: negative tug, about minus zero point one ish—call it a slight downward pull.

She gestures wide—positional first wave: solid positive lift, roughly zero point eight ish—like a firm upward shove.

Sum rough: lands around zero point eight ish, feels steadier, pasta finding its feet.

Next embedding piece: small positive kink, zero point zero ish—barely there nudge.

Her hand waves—positional second: decent rise, about zero point five ish—like half a push upward.

Glue it: roughly zero point six ish. Curve's settling, less wobble.

Third embedding: sharp negative edge, minus zero point one ish—jagged dip.

She signs quick—positional third: faint whisper, zero point zero ish—almost nothing.

Sum: dips to about minus zero point one ish. Sculpture sags low here, but clings—trust the shape's coming.

Last for the row: smooth positive bit, zero point one ish—gentle lift.

She gestures strong—positional fourth: full wave, one point zero ish—clean strong surge.

Result: climbs to about one point one ish. Snap—row locks solid. 'said' knows it's second. In your mind's eye the macaroni stands clearer: rough arch, word and place fused, no perfect lines but real form.
One row fused. Forty-ish more for ten tokens. Hands shake light, caffeine tailing off, eyes softening at edges. Chill seeps into joints.
Corner glance.
Assistant there. Charcoal turtleneck swallowing yellow light, face porcelain cut sharp. Hair tight back, eyes predatory still. No blink. Ozone cuts must like vent blast. Too crisp for this dusty sweat—8K against my grainy grind.
Yellow Post-it slash on oak. Pen bleeds: Element-wise sum. M_emb plus M_pos equals X. Four decimals later if needed.
Stand. Knees crack. Three steps. Air drops colder near her. Slide stacks and note.
Fingers long, clean lethal—draw papers in smooth. Hydraulic. Scent: cold copper post-rain, teeth twinge.
Turn back. Guilt nips neck—should earn every nudge through friction. She's the offload.
Swish quick. Fresh white sheet in light circle. Grey graphite—Courier fake, hand perfect. No smudge.
Look.
INPUT MATRIX X. Sequence ten. d_model four.
Row zero—'look' plus zero. First value: zero point zero ish—tiny positive flicker.
Row one—'said' plus one. First value: zero point eight ish—solid upward tilt.
Rest ten rows clean. Forty sums spot-on. Columns straight.
Paper cool touch.
This is X. Vectors fused. 'Look' owns front. 'Said' owns next. Macaroni sculpture complete—clumsy build, solid shape.
Glance her way. Wall stare. Still. Distractor. Accelerator. Efficiency bites.
Input set.
Next: Rainbow Ream. Split X. Queries. Keys. Values.
Black box hums stronger
……
CHAPTER 9: THE RAINBOW REAM – Color-Coding the Soul
It's 4:02 p.m., January 14, 2026. The silence in the auditorium is heavy, the kind that presses against your eardrums like deep water. The rain has stopped, but the dampness has seeped inside, making the air smell of wet wool and old books. The bare bulb overhead hums, a dying filament struggling against the gloom.
On the oak desk, the Input Matrix sits completed. Ten rows of fused data. But looking at rows of floating-point numbers is blinding. It’s noise. 0.037, 0.918... they mean nothing to the gut. To see what’s really happening, I have to stop thinking in digits and start thinking in tension.
I sweep the center of the desk clear. I need to see the dimensions. Not as columns on a page, but as physical things I can touch. I need to build a loom.
I raid the drawers. I dig into my lunch bag. I scavenge the room.
I tape four distinct lines across the oak surface to represent our four dimensions ($d_{model}=4$).
Dimension 0 (The Shoelace): I unthread a brown shoelace from my spare boot in the corner. I tape it taut. It’s frayed, soft, cottony. Sliding a bead on this is silent. No friction. This is the Semantic channel—soft meaning.

Dimension 1 (The Piano Wire): I scavenge a length of steel wire from the back of the typewriter mechanism. I pull it singing tight. It’s cold, sharp, dangerous. This is the Grammatical channel—rigid rules.
Dimension 2 (The Spaghetti): I find a piece of uncooked spaghetti from the box of "emergency rations" in the bottom drawer. I snap it to length. It’s brittle. If I push too hard, the dimension snaps. This is the Tone channel—fragile context.
Dimension 3 (The Phone Cable): I dig out an old white phone charger cable. Rubberized, grippy, tangled. Moving a bead on this takes effort; it fights you. This is the Positional channel—the drag of time.
I thread a heavy hexagonal nut onto each one to act as the "bead."
This is our vector space. A vector isn't a list of numbers anymore; it's a shape across this junk loom.
If I look at our first token, "Look":
* On the Shoelace, the nut slides lazily right.
* On the Piano Wire, the nut zips far right with a metallic shing.
* On the Spaghetti, the nut hovers nervously in the center.
* On the Phone Cable, I have to drag the nut hard against the rubber friction to the far end.
A jagged, tactile zigzag. That shape is the word "Look."
But the Transformer can't just look at this shape once. It needs to look at it three different ways. It needs to break this single physical reality into three distinct spirits.
New concept. Thrice rule. Query, Key, and Value.
First—the crisp dictionary definition:
The Self-Attention mechanism projects the input vector into three distinct subspaces: the Query (what the token is looking for), the Key (what the token identifies as), and the Value (the information the token actually passes along). These projections are created by multiplying the input by three separate learnable weight matrices: W_Q, W_K, and W_V.
Second—the functional warmth:
Think of a library filing system.
The Query is the note you hand the librarian: "I am looking for a book on 1940s primers."
The Key is the label on the spine of every book on the shelf: "History," "Fiction," "1940s." We match the Query to the Key to find the right book.
The Value is the book itself—the content inside that you actually take home once you've found the match.
Our model does this for every single word, all at once. "Look" asks (Query), "Look" advertises itself (Key), and "Look" offers its meaning (Value).
Third—the casual familiarity:
We’re just splitting our personality. Every word gets to play three roles.
The Red Query: The seeker.
The Blue Key: The badge.
The Green Value: The payload.
We take our jagged shape on the loom, run it through three different color filters, and get three different versions of itself.
I open the bottom drawer again. It groans. I pull out the stacks of construction paper I’ve been saving.
Crimson Red.
Deep Cobalt Blue.
Forest Green.
I place the colored stacks on the desk. The colors vibrate against the brown oak.

Red for Queries (Q).
Blue for Keys (K).
Green for Values (V).
Now, I need the filters. The Weight Matrices.
We have an input of 4 dimensions (Shoelace, Wire, Spaghetti, Cable).
We are projecting down to 2 dimensions for the attention heads.
So the matrices act as converters: they take the tension from our four junk lines and crush it down into two new, purely mathematical lines.
I turn to the Assistant. She is sitting perfectly still in the corner, the Input Matrix sheet resting aligned on her table.
"I need the weights," I say. My voice sounds rusty. "He initialization. 4×2 matrices. Red, Blue, Green."
She doesn't turn. She reaches for the colored paper.
Red. Blue. Green.
She slides them into the IBM wheel printer.
Whirr-clack-clack-clack.
The machine fires like a machine gun. Violent. Precise.
She stands and walks to the desk, placing the three colored sheets down.
The scent of ozone and copper hits me again. She retreats.
I look at the sheets. I don't read the numbers—I read the intent.
The Red Sheet (W_Q): This grid is aggressive. I see large negative numbers in the Phone Cable row. This matrix hates the phone cable. It wants to flip that friction around. It ignores the Shoelace entirely (zeros). It's a filter designed to strip away the soft meaning and focus on the hard position.
The Blue Sheet (W_K): This one is different. It emphasizes the Piano Wire. It wants to match based on grammar.
The Green Sheet (W_V): This is balanced. It wants to keep a little bit of everything to pass forward.
I line them up on the desk.
The Junk Loom, currently holding the shape of "Look."
Next to it, the Red Lens.
Next to that, the Blue Lens.
And the Green Lens.
The desk is a spectrum now. The grey gloom of the room can't touch this. We have the raw materials to start the engine.
But the nuts haven't moved yet. We've just loaded the gun.
To get our first Query vector—to actually see what "Look" is searching for—I have to manually crush the shape on the loom through the Red Sheet. I have to take the Shoelace value, the Wire value, the Spaghetti value, and the Cable value, and multiply them against the Red grid.
My hands twitch. The Sharp adding machine is waiting.
The Assistant has done the setup. She provided the matrices. But the act of projection? That's mine. I need to feel the compression. I need to feel the 4 dimensions snap down into 2.
I look at the Spaghetti dimension. It looks fragile. I hope it holds when I start the math.
"Head One," I whisper to the empty room. "Let's see what you're looking for."
The motor hums. Whiiine.
The friction begins.
……

OK, received. We are keeping the "Junk Loom" metaphor because it is tactile and perfect for audio visualization, but we are stripping out the recitation of the raw floats. The listener doesn't need to know it's 0.037; they need to know the shoelace is loose.
We will focus on the friction of the math, not the arithmetic itself.
Here is Chapter 10, revised for narrative flow, sensory depth, and the Thrice Rule, without the number-crunching fatigue.

CHAPTER 10: LINEAR PROJECTIONS – Calculating Queries for Head One
Setting: The desk is a wreck. 4:18 p.m. Rain drumming harder now. The "Input Loom" is rigged across the center—a shoelace, a piano wire, a spaghetti noodle, and a phone cord stretched tight.
(Sound: The heavy, rhythmic chug of the electric adding machine idling. The wet slap of rain against the window.)
I stare at the four strings stretched across the oak desk. This is my "Input Loom." It's a hack, a piece of desperation engineering to make sense of the invisible. We need to turn the raw token for "Look" into a Query. We need to find out what "Look" is asking for.
To do that, we have to run these four strings through the Red Matrix. We have to perform a Dot Product.
(Sound: Sharp intake of breath, tired.)
New term. Let's nail it down before the math drowns us.
First: The Definition.
The dot product is the sum of the products of the corresponding entries of two sequences of numbers. You multiply the first pair, the second pair, the third, and then you add all those results together into one single number.
Second: The Function.
Think of it as a compatibility check. A filter. I'm laying the reality of the input over the opinion of the weights. If they both push in the same direction, the signal gets loud. If they disagree—one positive, one negative—they cancel each other out. It tells us how much the input aligns with the lens we're looking through.
Third: The Grind.
It's just "multiply, multiply, multiply... sum." It's the rhythm of the adding machine. It's a grinder that chews up two lists of numbers and spits out a single verdict.
(Sound: I crack my knuckles. Dry, popping sound.)
I pick up the Red Sheet—the Query Matrix. It has two columns. Two different lenses to view the word "Look" through. We're doing the first one.
I look at the Loom.
Dimension Zero: The brown shoelace.
I slide the hex nut just a fraction to the right. The semantic value is low here. The shoelace is frayed, soft, creating zero friction. I check the Red Matrix. It doesn't care either. A small number times a small number.
(Sound: Quick, dismissive tick-tick on the machine.)
Barely worth the ink.
Dimension One: The steel piano wire.
This is the grammar channel. The nut is slid far to the right—high activation. The wire sings with tension. But the Red Matrix? It offers a negative weight. A rejection.
The input screams "Yes," but the lens says "No."

(Sound: A longer, grinding whiiine of the machine calculating a negative.)
The machine eats the conflict and spits out a subtraction. The signal is being suppressed.
Dimension Two: The dry spaghetti noodle.
I have to be gentle. The nut is barely moved. The tone is subtle here. The Red Matrix actually loves this dimension—it has a huge positive weight waiting—but my input is weak. It's a tragedy of potential. The lens is wide open, but the signal just isn't there to fill it.
(Sound: A soft, single tick.)
Wasted potential.
Dimension Three: The white phone charger cable.
This is position. The rubber is grippy, tangled. I've dragged the nut all the way to the end. It fights me. High friction. Massive input value.
I look at the Red Matrix. It hits back with a massive negative weight.
This is the main event. A huge positive force from the input colliding with a massive negative wall from the weights. The numbers are big. The machine is going to feel this one.
I punch it in. The keys feel stiff.
(Sound: The machine labors. WHIIINE-chug-chug-chug. It sounds like a car turning over in the cold.)
The negative weight crushes the input. The phone cable wins the argument. The previous three dimensions—the shoelace, the wire, the pasta—they don't matter. This one fight decided the war.
I hit the total key.
(Sound: KA-ka-ka-ka we-we-we ka-ka. The paper tape lurches forward.)
I tear off the strip. The number is negative. A deep, heavy negative.
I pick up the crimson fountain pen. The ink looks like dried blood in this light. on a fresh slip of paper, I write the result.
But we aren't done. That was just half the story. The Query head has two dimensions. We've compressed four strings into one number. Now we need the second coordinate.
I don't move the nuts on the loom. The word "Look" hasn't changed. The shoelace is still loose, the wire still tight.
But I shift my eyes to the second column of the Red Matrix. A new lens.
This time, the lens hates the shoelace, loves the wire, ignores the pasta, and is mild about the phone cable.
I go again. The rhythm takes over.
(Rhythmic, almost hypnotic)
Input against Weight.
Shoelace against Column.
Wire against Column.
(Sound: Tick-tick-whiiine.)
Spaghetti against Column.
Phone cable against Column.
(Sound: Tick-tick-whiiine. KA-ka-ka-ka we-we-we ka-ka.)
A second number. Positive this time.
I write it down next to the first one.
[-1.01, 0.42]
I cap the pen.
(Sound: Sharp click of the pen cap.)
I look at those two numbers. That is the Query.

We took the tactile mess of the loom—the friction, the tension, the fragility—and crushed it through the Red Matrix. We distilled "Look" down to these two coordinates.
This vector is a hunter. It is now stalking through the high-dimensional space, looking for a Key that looks like [-1.01, 0.42]. It doesn't want the word "Look." It wants a word that answers the question "Look" is asking.
One token down.
We still have to calculate the Key and the Value for this token.
And then... we have to do it for the other nine.
I look over at the IBM Wheelprinter in the corner. The "Assistant." It sits silent, mocking me with its speed. I could feed the tape in there. It could do the rest of the batch in seconds.
Not yet. The first one has to be manual. You have to feel the weight of the phone cable before you're allowed to automate it.
(Sound: I exhale, a long, shaky breath.)
Reset the machine. Clear the tape.
Next projection.
……
.....

CHAPTER 11: LINEAR PROJECTIONS – Calculating Keys for Head One

The rain has picked up again, steady drumming now, like someone's tapping Morse on the glass with a dull spoon. Four-eighteen has slipped into four-thirty-something; the yellow pool under the bare bulb feels smaller, meaner, as though the room is shrinking around the oak desk. My knuckles are raw from the adding machine keys. The shoelace (dimension zero) hangs a touch looser than before; the piano wire (dimension one) still sings when I brush it. Spaghetti (two) looks ready to crack if I breathe wrong. Phone cable (three) still fights every nudge of the hex nut.

We've hunted with the red lens. Now the word "look" must advertise itself—must declare what sort of match it can offer the rest of the sequence. Time for the blue lens. The Key.

New machinery, first proper appearance in our little opera, so we stop and give it the full thrice treatment before the fingers start dancing again.

First—the crisp, dictionary definition: A Key in self-attention is one of three learned projections of the input vector; it represents how the current token identifies itself so other tokens' Queries can decide how strongly to attend to it. Computed by multiplying the input vector by a dedicated weight matrix W_K.

Second—the functional warmth: The Key is the badge every word wears on its chest in the library of the sequence. While the Query asks "Who can help me right now?", the Key answers "This is who I am, this is what I've got." The dot product between a Query and a Key produces the raw compatibility score—how well this particular badge answers that particular question.

Third—the casual familiarity: Think of the Key as the price tag the word slaps on itself. Not what it's searching for, but what it's willing to sell. We run the same loom through the blue filter and see what price "look" is quoting.

I slide the blue construction-paper sheet forward until it overlaps the red one at the edges. The cobalt vibrates against the crimson like rival gang colours under sodium light. Two columns again—our head is keeping things tight, two dimensions per projection so the attention math stays sane on paper.

The loom hasn't moved. "Look" is still strung the same way: shoelace slack, piano wire taut, spaghetti timid, phone cable dragged hard against the rubber grip. The shape is fixed until the next token arrives.

I reset the adding machine tape—rip the old strip free with a satisfying tear, feed fresh white ribbon through the guides. Purple ink smells faintly of almonds and solvent. Motor hums awake with that familiar low drone.

First coordinate of the Key. Column one of the blue matrix.

I start at the shoelace. Semantic channel. Nut barely shifted—low value. Blue lens gives it a modest positive weight. Small times small. Tick-tick on the keys. Barely registers. Whiiine as I clear.

Piano wire next. Grammar channel screaming high. Nut slid far right, wire singing with tension. Blue matrix loves this one—large positive weight waiting. Positive times positive. The machine chugs longer here. WHIIINE-chug-chug. A decent positive contribution piles up.

Spaghetti. Tone channel. Nut hovering nervous in the middle. Fragile. Blue lens is lukewarm—small positive weight. Weak input times weak weight. Tick. Almost nothing.

Phone cable. Position channel. Nut dragged to the bitter end against rubber friction. High input value. Blue matrix hits back with a negative coefficient—sharp, deliberate. Positive meets negative. They fight. The machine groans under the opposition. WHIIINE … chug … chug-chug. The negative wins by a nose. The partial sum tips downward.

I punch the equals key. KA-ka-ka-ka we-we-we ka-ka. Tape lurches forward. Purple verdict blooms.

I tear the strip, glance at the number—solid positive, but tempered by that last collision. I reach for the cobalt fountain pen (ink darker, almost navy in this light) and write the first coordinate on a fresh slip beside the red Query numbers.

Now the second column of blue. New lens, same loom.

Shoelace again—still slack. Blue column two ignores it almost completely. Zero weight. Tick. Nothing.

Piano wire—still taut. This time the weight is negative and large. High positive input meets large negative opinion. They cancel hard. The machine sounds almost angry—WHIIINE-chug-chug-chug—like it's trying to lift something too heavy.

Spaghetti—timid. Weight small positive. Weak sauce. Tick.

Phone cable—still fighting the rubber. Input high. Weight here is small negative. High positive meets small negative. Net positive, but restrained.

Equals. KA-ka-ka we-we-we ka-ka.

Tape advances. I tear it free, read the second coordinate. Smaller than the first, but positive.

So there it is. The Key for "look" in head one: [some positive, smaller positive]. Not aggressive like the Query. Not hunting. Simply stating presence. A calm, confident price tag: "Here I am, and I'm worth this much attention if you need grammar or position."

I lay the blue slip beside the crimson one. They stare at each other like old adversaries who respect the game.

The loom hasn't budged. "Look" still strung the same—same tensions, same frictions. But now it wears three faces:

- Red Query: the hunter, leaning hard negative on position, asking an urgent question.
- Blue Key: the badge, balanced positive, quietly advertising.
- Green Value still waits in the wings—we'll get to the payload soon enough.

I lean back. Chair groans under me. Shoulders locked tight from hunching. The Go stones in the fishbowl click faintly when I shift—cool black touching cool white. The hundred-sided die hasn't rolled yet. Not chaos time.

The adding machine motor idles lower now, almost purring, as though it knows we've only just begun the real violence.

I glance toward the corner. The IBM wheel printer sits silent, the "Assistant" motionless beside it. She could hammer out the Keys for the other nine tokens in under ninety seconds. Whirr-clack-clack-clack. Perfect Courier. But no. Not yet. The first Key had to be felt—the piano wire singing against the negative weight, the phone cable fighting the rubber, the cancellation tasted through the tape's purple bloom.

One token fully projected into two of its three roles. Nine more waiting.

The rain drums harder. The bulb flickers once, then steadies.

I reset the tape again. Clear. Ready.

"Right," I mutter to the empty room. "Let's see what price 'said' is quoting."

The blue lens stays where it is. The loom waits.

Friction resumes.
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CHAPTER 12: LINEAR PROJECTIONS – Calculating Values for Head One

Four-forty-one p.m., January fourteenth, twenty-six. The rain has settled into a low, constant hiss against the window—like the room itself is exhaling through clenched teeth. The bare bulb overhead has taken on a deeper amber cast; shadows from the junk loom stretch long and thin across the oak, making the shoelace look almost threadbare, the piano wire gleam like a drawn blade. My shirt collar sticks to the back of my neck. Caffeine crash is creeping in, but the adding machine motor still idles steady, patient as ever. Purple tape coiled, ready for the next verdict.

The green construction paper waits in the stack—forest deep, almost black in this light. I slide it forward until it overlaps both red and blue at the corners. The three sheets together form a tricolor flag of intent: seek, identify, deliver. We've asked. We've advertised. Now we pay up.

New role in the machinery, first proper time through the grinder, so we halt and give Value the full thrice treatment before the keys start snapping again.

First—the crisp, dictionary definition: The Value in self-attention is the third learned projection of the input vector; it represents the actual content or information that will be passed forward to the current position if other tokens decide to attend to it. Computed by multiplying the input vector by a dedicated weight matrix W_V.

Second—the functional warmth: While the Query hunts and the Key advertises, the Value is the cargo hold. It's what gets delivered when a high attention score is granted. Every token prepares a little parcel of itself—meaning, grammar hints, positional flavor—and offers it up. The attention weights then decide how much of each parcel ends up blended into the final output for this position.

Third—the casual familiarity: Think of the Value as the wallet the word opens when someone pays attention. Not the question it asks, not the price tag it shows—just the cash, the goods, the stuff that actually changes hands. We run the loom through the green filter one last time and see what "look" is willing to hand over.

The loom hasn't shifted an inch since we strung "look." Shoelace slack (semantic soft), piano wire taut (grammar rigid), spaghetti timid (tone fragile), phone cable dragged hard against rubber grip (position fighting every inch). Same jagged shape. Same four tensions. Different lens.

I reset the adding machine tape—rip the old strip free with a quick tear that echoes sharper than it should in the damp air. Fresh white ribbon threads through. Motor hums awake, low and content. Green fountain pen uncapped; ink smells faintly of moss and iron.

First coordinate of the Value. Column one of the green matrix.

Start at the shoelace. Semantic channel. Nut barely moved—low input. Green lens gives it a decent positive weight. Small times decent. Tick-tick. A modest positive contribution slides onto the running total.

Piano wire next. Grammar screaming high. Nut far right, wire singing tension. Green matrix offers a strong positive coefficient here—loves the grammar signal. High positive meets high positive. The machine chugs with satisfaction. WHIIINE-chug-chug-chug. A healthy pile accumulates.

Spaghetti. Tone channel. Nut hovering nervous. Fragile input. Green lens is gentle—small positive weight. Weak times small. Tick. Barely registers, but it adds a whisper.

Phone cable. Position channel. Nut dragged to the bitter end. High friction input. Green matrix hits back with a moderate negative weight—not crushing, but firm. High positive meets moderate negative. They wrestle. The machine groans under the opposition. WHIIINE … chug … chug-chug. The positive just edges it out. Net positive, but bruised.

Equals key. KA-ka-ka-ka we-we-we ka-ka. Tape lurches. Purple ink blooms crisp.

I tear the strip, glance at the number—solid positive, warmer than the blue Key's offering. I reach for the forest-green fountain pen and write the first coordinate on a fresh slip beside the red and blue numbers.

Second column of green. New lens, same loom.

Shoelace again—still slack. Green column two gives it a small negative weight. Low input meets small negative. Tick. Tiny subtraction.

Piano wire—still taut. This time the weight is large positive again. High input meets large positive. They reinforce hard. WHIIINE-chug-chug-chug. The machine sounds almost pleased.

Spaghetti—timid. Weight moderate positive. Weak input gets a gentle boost. Tick-tick. Small but kind addition.

Phone cable—still fighting rubber. High input. Weight here is small positive. High positive meets small positive. Net strong positive.

Equals. KA-ka-ka-ka we-we-we ka-ka.

Tape advances. I tear it free. Second coordinate reads larger than the first—decidedly positive.

So the Value for "look" in head one settles as [decent positive, larger positive]. Not hunting like the Query. Not merely advertising like the Key. Offering. Warm, willing to be taken. A parcel packed with grammar strength and positional flavor, softened by a touch of semantics and tone. If any later Query decides this "look" is worth attending to, this is the blend that flows forward.

I lay the green slip beside its red and blue siblings. Three colored rectangles now stare up from the oak: crimson hunter, cobalt badge, forest cargo. Together they are the full triune personality of the first token—projected, compressed, ready.

The loom still holds the shape of "look." Four strings, four nuts, four tensions unchanged. But the word has been split three ways and offered back to the sequence in three distinct costumes.

I lean back. Chair creaks under the weight of accumulated fatigue. Shoulders locked like rusted hinges. The fishbowl of Go stones sits cool and untouched to my right; black touching white in perfect silence. The hundred-sided die perches on top, green plastic catching the amber light like it's waiting its turn to introduce true randomness later—maybe at epoch boundaries, maybe when overfitting threatens to choke us.

The adding machine motor drops to a softer idle, almost purring now. Purple tape coiled patient. The rainbow ream in the corner—red, blue, green already used, yellow for feed-forward and violet for gradients still waiting their chapters—seems to breathe a little easier.

I glance toward the corner. The IBM wheel printer sits silent. The Assistant motionless beside it, charcoal turtleneck absorbing the light, posture impossibly straight. She could batch-project the Values for the remaining nine tokens in a minute flat. Whirr-clack-clack-clack. Perfect columns. But no. Not for this one. The first Value had to be felt—the piano wire reinforcing itself through the green lens, the phone cable softened by moderation rather than crushed, the small kindness the spaghetti received when it was weakest.

One token fully projected. Three faces prepared. Nine more tokens still wearing only their fused input vectors, waiting to be split.

The rain hisses steady. The bulb flickers once—warning or just age—then steadies.

I reset the tape again. Clear. Fresh.

"Right," I mutter, voice gravel from too many quiet hours. "Let's see what 'said' is willing to hand over."

The green lens stays where it is. The loom waits, strings taut.

Friction resumes. The black box doesn't just hum now—it begins to sing, low and dangerous.

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CHAPTER 13: SCALED DOT-PRODUCT ATTENTION – Raw Scores and Division

Four-fifty-eight p.m., January fourteenth, twenty-six. The auditorium feels smaller, the air thicker—like the rain outside has seeped through the mortar and is pressing the walls inward. The bare bulb flickers once, twice, then steadies with a faint buzz that matches the low throb of the adding machine motor. My fingers ache from the keys; ink smudges the webbing between thumb and forefinger. The junk loom stretches across the oak: shoelace slack, piano wire singing, spaghetti trembling, phone cable fighting. Three colored slips—crimson, cobalt, forest—lie in a neat row beside it. Query, Key, Value for "look." Head one. The first token is fully armed.

Now the machinery turns inward. Self-attention begins. Every token must speak to every other token—including itself—through these projections. We need scores. Raw compatibility numbers. How much does this Query want that Key? How loudly does one badge answer another's hunt?

New mechanism, first proper appearance, so we stop dead and give scaled dot-product attention the full thrice treatment before the violence starts.

First—the crisp, dictionary definition: Scaled dot-product attention computes raw attention scores by taking the dot product of the Query vector and the Key vector, then dividing the result by the square root of the Key's dimension (here $\sqrt{2}$) to prevent the scores from growing too large as dimensions increase. These raw scores indicate how much attention one position should pay to another.

Second—the functional warmth: The dot product measures alignment—how parallel the Query arrow points to the Key arrow in vector space. High positive means strong match; negative means opposition; near zero means indifference. Scaling by $\sqrt{d_k}$ ($d_k=2$ for our head) keeps the values from exploding when we later softmax them—without it, the gradients would vanish or blow up, and the adding machine would choke on numbers too big or too small to handle cleanly.

Third—the casual familiarity: It's just "how similar are these two vectors?" multiplied out, then shrunk a touch so the numbers stay sane for the next step. Dot product shouts the raw attraction; scaling whispers "easy now, don't get carried away."

We have only one token projected so far—"look"—so for the moment the attention matrix is a lonely one-by-one affair. But the principle holds: Query dot Key, scale, record. Later, when the full sequence is projected, we'll build the full square grid of scores—ten rows by ten columns for our primer corpus max.

I clear the adding machine tape—rip the old strip free, the tear louder than it should be in the damp quiet. Fresh ribbon threads through. Purple ink smells sharp. Motor hums awake.

First: the raw dot product between "look"'s Query and its own Key.

From earlier slips:

Query (red): let's call it Q = [-1.01, 0.42] (the hunter's lean)

Key (blue): K = [positive decent, positive smaller] — say roughly [0.78, 0.31] for the badge's calm offer

No need to re-punch every digit; we've got the colored slips laid out like playing cards. I read them aloud to the empty room, voice gravel:

"Query: negative one point oh one, point four two."

"Key: point seven eight, point three one."

Now the dot product: first components multiply, second components multiply, then sum.

I punch the first pair into the Sharp: negative one point oh one times point seven eight.

WHIIINE-chug-chug. Negative contribution. The machine sounds almost disappointed.

Second pair: point four two times point three one.

Tick-tick. Small positive.

Equals. KA-ka-ka-ka we-we-we ka-ka.

Tape lurches. Purple blooms a modest positive number—small, because the negative leg dragged hard, but not zero. The Query and Key for the same token are not perfectly aligned; there's tension even in self-regard.

Now scale it. Divide by square root of two—roughly one point four one four.

I punch the raw score, then divide.

WHIIINE … chug … equals.

KA-ka-ka-ka we-we-we ka-ka.

Smaller still. A gentle number now, neither screaming nor whispering. Just honest attraction from self to self.

I reach for a fresh white slip—no color yet, this is raw—and write the scaled score in black fountain pen. Ink bleeds faintly at the edges.

This is the (1,1) entry of our eventual attention matrix: how much "look" attends to itself in head one. Not overwhelming—there's friction even when talking to its own reflection—but positive. Enough to matter.

In a full sequence the process repeats: each Query dots every Key, row by row. Ten Queries, ten Keys, one hundred dot products. Each scaled by the same $\sqrt{2}$. A full grid of compatibility whispers. Then—and only then—do we mask the future (next chapter) and softmax the rows (chapter after) to turn raw scores into proper probabilities.

For now, though, we have only this single lonely score. A proof of concept. The machinery works. The dot product speaks. The scaling tames it.

I lean back. Chair groans. Shoulders locked tighter than the piano wire. The fishbowl Go stones click softly when I shift—cool black against cool white. The hundred-sided die gleams patient on top, still waiting its chaos turn.

The adding machine idles lower, almost content. Purple tape coiled. The rainbow ream in the corner—red, blue, green used, yellow and violet still pristine—seems to breathe easier now that attention has begun to stir.

I glance toward the corner. The IBM wheel printer silent. The Assistant motionless, charcoal turtleneck drinking the light, posture straight as rebar. She could punch the full ten-by-ten grid in minutes once all projections are done. Whirr-clack-clack-clack. Perfect rows. But not yet. This first self-dot had to be felt—the negative drag of the first component fighting the modest lift of the second, the scaling pulling it back from boast to honesty.

One entry in the attention matrix. Ninety-nine more waiting.

The rain hisses steady. The bulb buzzes faint.

I reset the tape. Clear. Fresh.

"Right," I mutter. "Let's project 'said' next—Query, Key, Value—before the room decides we've had enough for one dreary afternoon."

The loom waits, strings taut.

The black box isn't just singing now. It's beginning to listen.

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CHAPTER 14: MASKING THE FUTURE – Blinding the Model to Unseen Tokens

Five-twelve p.m., January fourteenth, twenty-six. The auditorium has gone properly dim; the bare bulb fights a losing battle against the January gloom pressing in from the single tall window. Rain still hisses steady, but softer—like the room itself is trying to hush what comes next. My eyes burn from the yellow light and too many slips of paper staring back. Fingers stiff; ink has dried in thin black lines across the knuckles. The junk loom holds "look" frozen: shoelace slack, piano wire taut, spaghetti trembling, phone cable dragged hard. Three colored slips lie beside it—red hunter, blue badge, green cargo. One lonely scaled score sits on white paper: the one-one entry, self-attention's first confession.

Before we turn those raw scores into probabilities, before softmax can do its gentle redistribution, we must enforce causality. The model cannot know what comes after. In training, every position must predict the next without cheating by glancing ahead. Enter the mask.

New piece of the engine, first proper turn through the grinder, so we stop and give causal masking the full thrice treatment before the tape advances again.

First—the crisp, dictionary definition: Causal masking, or future masking, in self-attention sets all scores in the attention matrix that correspond to future positions—where the column index is greater than the row index—to negative infinity before applying softmax. This forces the attention weights for any position to sum only over positions one through itself, preventing information from later tokens leaking backward during training.

Second—the functional warmth: Without the mask, the model would see the entire sequence at once—cheating like a student reading tomorrow's exam answers today. The mask enforces autoregressive behavior: each token can only attend to what has come before it and itself. It

preserves the arrow of time we injected with positional encodings. No shortcuts. No spoilers. The adding machine only sees what's already happened.

Third—the casual familiarity: It's just slapping a big "nope" on every cell to the right of the diagonal in the attention grid. Future scores get drowned in negative infinity so that when softmax hits, their weights drop to absolute zero. Clean. Brutal. Fair.

Right now our attention matrix is still a sad one-by-one square—just that single self-score for "look." No future to mask yet. But the principle must be laid down before we project the full sequence and build the real grid. Ten tokens eventually means a ten-by-ten matrix of scaled dot products. The upper triangle—everything above the main diagonal—must die.

I reach for a fresh sheet of bright white copy—fourteenth layer cluttering the oak now—and thread it into the IBM typewriter. Platen grips with a soft crunch. Index three times—click-thunk, click-thunk, click-thunk. Header slow, deliberate:

CAUSAL MASKING – ATTENTION MATRIX STRUCTURE (for sequence length ten maximum)
Before softmax: future positions (column greater than row) set to negative infinity

I type a skeleton grid—ten rows, ten columns, just outlines for now. No numbers yet. Just labels:

Row one (token one "look"): [score one-one, negative infinity, negative infinity, negative infinity, negative infinity, negative infinity, negative infinity, negative infinity, negative infinity, negative infinity]
Row two (token two "said"): [score two-one, score two-two, negative infinity, negative infinity, negative infinity, negative infinity, negative infinity, negative infinity, negative infinity, negative infinity]
...
Row ten: [score ten-one ... score ten-ten]

Each "negative infinity" I type in full—n-e-g-a-t-i-v-e space i-n-f-i-n-i-t-y—so the voice reads it clean and even. The buckling springs snap with every entry. Mistype a row label? Whirr—correction ribbon rises slow, white bottom lifting the error clean away like breath on glass. Retype. Perfect.

In the mind's eye this isn't just a grid on paper. It's a stone wall built across time. Position one can see only itself. Position two sees one and two. Position ten sees the whole bloody history up to ten—but never eleven, never tomorrow. The mask is the bricklayer's trowel slapping mortar on every future-facing gap. No light leaks through.

Later, when we have all ten rows of Queries and ten columns of Keys projected, we'll compute the full ten-by-ten scaled dot-product matrix. Then, for every cell where column index is greater than row index, we overwrite the raw score with negative infinity. Not a small negative. Not a gentle penalty. Absolute drowning. When softmax hits each row, those negative infinity terms become zero probability—vanished. The row sums only over the past and present.

I pull the sheet free with a rip that echoes too loud in the quieted rain. Lay it beside the colored slips and the lonely self-score. The mask grid stares up—mostly negative infinity, a thin diagonal of possibility snaking through the drowned futures.

The adding machine idles low, patient. Purple tape coiled. The rainbow ream in the corner—red, blue, green used for projections, yellow still waiting for feed-forward, violet for gradients—seems to lean in closer now that causality is locked.

I lean back. Chair creaks under accumulated hours. Shoulders knotted like old rope. The fishbowl Go stones sit cool and silent; black touching white in perfect balance. The hundred-sided die perches on top, green plastic catching the amber bulb like it's dreaming of the randomness we'll need when overfitting starts to bite.

I glance toward the corner. The IBM wheel printer silent. The Assistant motionless beside it—charcoal turtleneck absorbing what little light remains, posture straight as a ruled line. She could print full masked matrices for a dozen sequences in seconds once we have them. Whirr-clack-clack-clack. Perfect Courier. But not yet. The mask had to be felt first—typed slow, correction ribbon used once for a crooked label, the weight of all those drowned futures sinking into the paper.

Causality enforced. Time blinded to its own tail.

The rain hisses softer now, almost thoughtful.

I reset the tape. Clear. Fresh.

"Right," I mutter, voice rough from the long afternoon. "We've got the scores scaled and the future buried. Next we turn those drowned numbers into probabilities that actually sum to one."

The loom waits, strings still taut with "look." Nine more tokens yet to project.

The black box isn't just listening anymore. It's learning to forget what it shouldn't know.

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CHAPTER 15: SOFTMAX FUNCTION – Turning Scores into Probabilities

Five-thirty-three p.m., January fourteenth, twenty-six. The auditorium has slipped into proper evening shadow; the bare bulb overhead burns like a tired coal, throwing long fingers across the oak desk. Rain has eased to occasional drips—plink, plink—against the gutter outside. Inside, the air hangs heavy with ribbon solvent, warm machine oil, and the faint metallic tang of fountain-pen ink on my fingers. Shoulders locked solid. Eyes gritty. The junk loom still holds "look" in its four-string grip: shoelace slack, piano wire taut, spaghetti fragile, phone cable fighting. Three colored slips—red, blue, green—lie beside the white paper bearing our lonely scaled self-score. The mask grid looms nearby, mostly negative infinity, a thin diagonal of possibility snaking through.

We have raw attention scores—modest positive for self-attention in head one so far. But raw numbers aren't decisions. They need to be turned into weights: fractions between zero and one that add up to exactly one for each row. Enter softmax.

New operation, first proper pass through the machinery, so we halt and give it the full thrice treatment before the adding machine wakes again.

First—the crisp, dictionary definition: Softmax is a function that converts a vector of raw scores (logits) into a probability distribution by taking the exponential of each score, then dividing each exponential by the sum of all exponentials in the vector. This ensures all outputs are positive and sum to one.

Second—the functional warmth: It turns competition into cooperation. High raw scores get amplified exponentially—the winner takes most—but never everything. Low scores shrink toward zero. After masking, the future is already dead; softmax now redistributes what's left so each position knows exactly how much attention to pay to the past and present. No ties. No leftovers. Clean probabilities for weighted blending.

Third—the casual familiarity: It's the "make it fair but favor the strong" button. Raw score gets exponentiated—big ones balloon, small ones shrivel—then everything gets normalized so the row adds up tidy to one. Like pouring votes: the popular kid gets the lion's share, but the quiet ones still get a whisper.

For our single-token case, it's almost trivial—one score, one row of one. But the math must be felt before we scale to ten-by-ten.

I clear the tape—rip the old strip free, the tear sharp in the quiet. Fresh ribbon threads through. Purple ink smells of almonds and regret. Motor hums awake, low and steady.

Our lone scaled score for "look" attending to itself—let's call it roughly zero point six eight (from earlier dot-product and scaling; modest but positive).

First step: exponentiate it. e raised to zero point six eight.

I reach for the Standard Mathematical Tables again—spine cracked, pages vanilla-dusty—and flip to the exponential section. Finger traces the columns. e^0.6 ≈ one point eight two one, e^0.7 ≈ two point zero one three. Interpolate rough: call it about one point nine seven.

Punch it into the Sharp for precision—approximate is fine for feel, but we need the habit.

WHIIINE-chug-chug. Tape blooms one point nine seven (close enough; the machine doesn't argue).

Since it's a single-element vector after masking (only self remains), the sum of exponentials is just this one number.

Now divide: exp(score) divided by sum(exp) = one point nine seven divided by one point nine seven.

Equals one. Exactly.

KA-ka-ka-ka we-we-we ka-ka.

Tape lurches. Purple verdict: one point zero zero.

I tear the strip, lay it beside the scaled score. Attention weight for "look" to itself: full one hundred percent. No surprise—nothing else exists yet. But the ritual matters. The machinery has tasted exponentiation, tasted normalization.

In the full sequence, every row will look different. A row with multiple live scores (past and present) gets multiple exponentials—big ones dominate, small ones fade—then divide by their collective sum. After causal mask, no future leaks in; the probabilities flow only backward and inward.

I reach for a fresh white slip—no color needed yet—and write in black fountain pen:

Softmax example (single position):
Raw scaled score: ~0.68
exp(~0.68) ≈ 1.97
Sum of exps = 1.97
Softmax weight: 1.97 / 1.97 = 1.0

Later rows will be messier—multiple live cells, exponentials fighting for dominance, the mask ensuring only the left side plays. The winner gets the bulk of the attention; the losers get crumbs.

The adding machine idles softer now, almost satisfied. Purple tape coiled patient. The rainbow ream waits in the corner—yellow for feed-forward next, violet for gradients looming after.

I lean back. Chair groans under the day's weight. The fishbowl Go stones sit cool; black and white in silent agreement. The hundred-sided die gleams on top, patient for its chaos cue.

I glance to the corner. The IBM wheel printer silent. The Assistant motionless—charcoal turtleneck drinking the dim light, posture straight as justice. She could softmax a full ten-by-ten grid in a blink once we have it. Whirr-clack-clack-clack. Perfect rows of probabilities. But not yet. This first trivial softmax had to be tasted—the exponent lifting the modest score, the division locking it to one, the certainty of full self-attention in an empty room.

One probability vector complete. Ninety-nine more waiting.

The drips outside slow to almost nothing.

I reset the tape. Clear. Fresh.

"Right," I mutter, voice rough as old gravel. "We've got probabilities now. Next we blend the Values according to those weights—weighted sum time."

The loom waits, strings taut with "look." Nine tokens still unprojected.

The black box isn't just listening and forgetting. It's beginning to choose.

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CHAPTER 16: WEIGHTED SUM – Blending Values with Attention Weights

Five-forty-nine p.m., January fourteenth, twenty-six. The auditorium has gone quiet enough you can hear the bulb filament sigh every few seconds. Rain drips slow and thoughtful outside—plink … plink—like the world itself is counting heartbeats. Inside, the air clings: warm oil from the adding machine, faint almond bite of purple ink, metallic tang of fountain-pen residue on my thumbs. Shoulders feel like they've been bolted in place. Eyes raw. The junk loom still grips "look": shoelace slack across dimension zero, piano wire taut on one, spaghetti trembling on two, phone cable dragged hard against rubber on three. Three colored slips lie in a row—crimson Query, cobalt Key, forest Value. Beside them, the scaled self-score on white paper, the softmax weight of one point zero zero, the mask grid mostly negative infinity with its thin diagonal lifeline.

We have attention weights now. For this single position, the weight to itself is full—one hundred percent. No dilution yet. But the principle stands: multiply each Value vector by its corresponding attention weight, then sum them all up per row. The result is the attended representation for that position—blended information from everywhere the model decided to listen.

New step in the dance, first proper execution, so we pause and give weighted sum the full thrice treatment before the tape starts rolling again.

First—the crisp, dictionary definition: Weighted sum in self-attention computes the output for each position by taking a linear combination of all Value vectors, where the coefficients are the attention probabilities from the softmax step. For each row i, output_i equals sum over j of (attention_weight_i,j times Value_j).

Second—the functional warmth: This is where attention pays off. Instead of treating every token equally, the model pulls more from the tokens it deemed relevant (high weight) and almost nothing from the irrelevant ones (low or zero weight). After masking, only past and present contribute. The result is a new vector for each position—richer, context-aware—still in the head's dimension (two here), ready to be stitched back with other heads and projected forward.

Third—the casual familiarity: It's just "pay more attention to the good bits." Each Value gets multiplied by how much the model cares about it, then everything adds up. High-weight Values dominate the blend; low-weight ones fade to whispers. Clean math, honest outcome.

Our case stays trivial for now—one position, one live Value (its own green projection), one attention weight of one. So the weighted sum is simply the Value itself—no blending needed yet. But we walk the full path anyway; habit matters more than convenience.

I tear the old tape free—sharp rip echoing in the hush. Fresh ribbon threads through the Sharp. Purple ink smells sharp and ready. Motor hums awake, low contented drone.

Our forest Value for "look" in head one—call it roughly [one point two three, one point eight seven] from earlier green projections (decent positive first coordinate, larger positive second).

Attention weight to itself: one point zero zero.

Multiply: one point zero zero times the Value vector.

First coordinate: one times one point two three equals one point two three.

Punch it in. Tick-tick. WHIIINE-chug. No fight here—unity times anything is itself.

Second coordinate: one times one point eight seven equals one point eight seven.

Same rhythm. KA-ka-ka-ka we-we-we ka-ka.

Tape blooms the result: [one point two three, one point eight seven]

Exactly the original Value. No change. No averaging. Full self-copy.

I tear the strip, lay it beside the softmax weight. Write on a fresh white slip in black fountain pen:

Weighted sum (position one, head one):
Attention weight to self: 1.0
Value_self × 1.0 = [~1.23, ~1.87]
Output for position one: [~1.23, ~1.87]

Trivial today. But imagine the full grid: ten positions, each row with its softmax weights summing to one, most zeros from the mask, a handful of live fractions pulling from earlier Values. Row two ("said") might pull eighty percent from "look"'s Value and twenty from itself. The output vector for position two becomes a weighted cocktail—mostly "look"'s green parcel, flavored by its own. Richer than raw input. Context baked in.

This head-one output vector—still two-dimensional—will soon concatenate with head-two's output, then project back to our original four dimensions. But that's later chapters. For now, the weighted sum has spoken: attention delivers.

The adding machine idles softer, almost purring. Purple tape coiled patient. The rainbow ream in the corner—yellow feed-forward and violet gradients still untouched—seems to settle, knowing the core loop is closing.

I lean back. Chair creaks under the accumulated hours. The fishbowl Go stones sit cool and silent; black and white stones touching in quiet agreement. The hundred-sided die perches on top, green plastic catching the dim amber glow, waiting for its moment when randomness might save us from collapse.

I glance toward the corner. The IBM wheel printer silent. The Assistant motionless beside it—charcoal turtleneck absorbing the failing light, posture straight as a ruled margin. She could hammer out weighted sums for a full ten-by-ten attention matrix in under a minute once everything's projected. Whirr-clack-clack-clack. Perfect columns of blended vectors. But not yet. This first trivial blend had to be felt—the multiplication clean and unchallenged, the sum unchanged because nothing else existed to pull from, the quiet certainty of self-attention in solitude.

One output vector complete for head one, position one. Ninety-nine more waiting across the sequence.

The drips outside have stopped entirely. Silence presses in.
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CHAPTER SEVENTEEN: CONCATENATION AND OUTPUT PROJECTION – Stitching Heads Back Together

Six-oh-seven p.m., January fourteenth, twenty-six. The auditorium has surrendered to full dark outside; the single tall window is now just a black mirror reflecting the bare bulb's amber struggle. No more rain—only the faint drip from the gutter, slow as regret. Inside, the air has thickened: warm oil from the Sharp, ribbon solvent clinging to my cuffs, the iron bite of green fountain-pen ink still wet on the last slip. Shoulders feel cast in lead. Eyes sting like they've been sanded. The junk loom holds "look" frozen across its four strings—shoelace slack, piano wire singing, spaghetti fragile, phone cable fighting rubber grip. Colored slips lie in neat row: crimson Query, cobalt Key, forest Value for head one. Beside them, the scaled score, the softmax weight of one, the weighted sum output vector still two-dimensional and identical to the original Value.

We have one head's output for position one: a two-dimensional blend [roughly one point two three, one point eight seven]. In the full machinery there would be a second head—parallel, independent—projecting the same input through different red, blue, green lenses, computing its own attention scores, its own weighted sum. Two separate two-dimensional outputs. Before they feed into the feed-forward layer, we must stitch them back into the original model dimension—four here.

New operation, first proper appearance, so we stop and give concatenation and output projection the full thrice treatment before the typewriter wakes.

First—the crisp, dictionary definition: Concatenation in multi-head attention joins the output vectors from all attention heads along the last dimension, producing a single longer vector per position. The output projection then applies a linear transformation—multiplication by a learned weight matrix W_O followed by optional bias—to map this concatenated vector back to the model's hidden dimension d_model.

Second—the functional warmth: Different heads learn to attend to different patterns—syntax in one, semantics in another, long-range dependencies in a third. Concatenation lets them pool their discoveries without forcing compromise during computation. The final projection mixes those pooled insights back into a coherent signal the rest of the transformer can use, preserving the original vector length for residual connections and layer stacking.

Third—the casual familiarity: It's like having two mates scout different pubs—one finds the good ale, the other spots the quiet corner—then you mash their reports together and run the whole thing through one last filter so it fits back in your notebook the same size it started. Heads work in parallel; concatenation glues; projection tunes the glue.

Our case remains skeletal—one head active, second not yet projected. So concatenation is trivial: just the single two-dimensional vector from head one. No second vector to join. But the shape must be prepared. In full flow: head-one output [two dims] next to head-two output [two dims] makes a four-dimensional concatenated vector. Then multiply by W_O (four by four matrix) to squeeze or reshape back to d_model equals four.

I reach for the rainbow ream—violet sheet this time, reserved for final projections and gradients later—and slide it under the typewriter platen. Index twice—click-thunk, click-thunk. Header deliberate:

MULTI-HEAD ATTENTION OUTPUT – CONCATENATION AND PROJECTION (head one only for now)

I type the skeleton:

Head 1 output vector (position 1): [~1.23, ~1.87]
Head 2 output vector: [not yet computed]
Concatenated: [~1.23, ~1.87, <head2_dim0>, <head2_dim1>]
Output projection: concatenated × W_O (4×4 matrix) → back to d_model=4

Since head two sleeps, I note in black fountain pen on a fresh white slip:

Current concatenated vector (partial): [~1.23, ~1.87] padded conceptually to four dims for projection prep.

The real work waits. Once head two is projected—same loom through different red, blue, green lenses—we'll have the full four-dimensional concatenation. Then the output projection matrix (four rows, four columns, learned weights) crushes it back to four. Dot products again—four input components against four output rows. The Sharp will hum through sixteen multiplications and four sums.

I pull the sheet free with a soft rip. Lay it beside the growing stack. The violet paper catches the bulb's amber like bruised twilight.

In the mind's eye: two scouts return from separate hunts. One brings a two-piece report. The other hasn't arrived. We tape the first report to a blank second sheet—concatenation—then run the whole page through one last lens (W_O) so it matches the size and shape of the original input vector. Residual connection can add it back cleanly later. Layer norm can tame it. Feed-forward waits downstream.

The adding machine idles low, patient. Purple tape coiled. The rainbow ream settles—violet now touched, yellow feed-forward next in line.

I lean back. Chair groans under the day's toll. The fishbowl Go stones sit cool; black and white in silent council. The hundred-sided die gleams on top, green plastic patient for the epoch when true randomness might break overfitting's grip.

I glance to the corner. The IBM wheel printer silent. The Assistant motionless—charcoal turtleneck drinking the last of the light, posture straight as a verdict. She could concatenate and project full multi-head outputs for the entire sequence in moments once both heads are live. Whirr-clack-clack-clack. Perfect four-dimensional vectors. But not yet. This partial stitch had to be felt—the lonely two-dimensional output from head one waiting for its twin, the conceptual padding to four, the promise of W_O waiting to mix them into something greater than the sum.

One position's multi-head output prepared—half-complete. Ninety-nine more positions, one more head per position, waiting.

The silence outside is complete now. No drips. Just the bulb's faint buzz.

I reset the tape. Clear. Fresh.

"Right," I mutter, voice scraped raw. "Head one is home. Time to string head two on the loom—new red, new blue, new green—and give 'look' its second face before the night claims us entirely."

The loom waits, four strings still taut.

The black box isn't just remembering what it chose. It's learning to speak with two voices at once.

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CHAPTER 18: LINEAR PROJECTIONS – Calculating Queries, Keys, and Values for Head Two

Six-twenty-one p.m., January fourteenth, twenty-six. The auditorium feels like it's holding its breath. The bare bulb overhead has settled into a steady, weary amber glow; shadows from the junk loom stretch long and thin across the scarred oak. No rain now—just the faint metallic creak of the building cooling. My collar sticks to the neck; ink has dried in fine black webs between fingers. The fishbowl Go stones sit cool and untouched; black touching white in silent patience. The hundred-sided die perches on top, green plastic catching the dim light like it's dreaming of the chaos we'll need when the model starts memorizing noise instead of signal.

Head one is done for "look"—Query hunted, Key advertised, Value offered, attention computed, weighted sum blended, partial concatenation waiting. Head two must now run parallel. Same input vector (fused embedding plus positional), different lenses. Different red, blue, green matrices tuned to catch patterns the first head missed—perhaps longer dependencies, subtler tone, different grammatical hooks. Parallel scouts. Independent hunts.

New head means new projections. We stop and remind ourselves of the machinery, though the thrice treatment stays light this time—familiar ground, just fresh colors.

First—the crisp reminder: Each attention head has its own independent set of learned weight matrices W_Q, W_K, W_V. These project the same input vector into separate subspaces, allowing different heads to specialize on different aspects of the relationships in the sequence.

Second—the functional warmth: Head one might have latched onto positional friction and grammar tension; head two could emphasize semantic softness or tone fragility. Parallel computation lets the model attend to multiple patterns at once without forcing a single compromise. More heads, more eyes—same input, richer understanding.

Third—the casual familiarity: It's like hiring a second mate to scout the same territory through different binoculars. Same hill, different lenses—one spots the pub, the other spots the shortcut. We run the loom through fresh crimson, cobalt, forest sheets and see what head two makes of "look."

I sweep the first set of colored slips aside—crimson, cobalt, forest for head one—and reach for the rainbow ream again. Fresh crimson sheet for Queries, head two. Deep cobalt for Keys. Forest green for Values. The colors vibrate against the oak like rival banners under sodium light.

The loom hasn't budged. "Look" still strung the same: shoelace loose (semantic soft), piano wire tight (grammar rigid), spaghetti timid (tone fragile), phone cable dragged hard (position fighting). Same tensions. Different filters.

I thread a fresh white slip into the typewriter first—header clean:

HEAD TWO PROJECTIONS – QUERIES, KEYS, VALUES (position 1 "look")

Then reset the Sharp tape—rip the old strip free, fresh ribbon through. Purple ink blooms ready. Motor hums awake.

Start with Queries, head two. Crimson sheet number two.

First coordinate—column one of the new red matrix.

Shoelace: low input, loose. Head-two crimson gives it a modest positive weight this time. Small times modest. Tick-tick. Gentle positive contribution.

Piano wire: high tension. New matrix offers a negative coefficient—rejection. High positive meets negative. WHIIINE-chug-chug. Signal suppressed.

Spaghetti: timid hover. Weight small positive. Weak input gets a small lift. Tick.

Phone cable: high friction. Matrix hits with large positive—amplification. High input meets large positive. The machine labors happily. WHIIINE … chug-chug-chug. Strong positive piles up.

Equals. KA-ka-ka-ka we-we-we ka-ka.

I tear the tape, read the first coordinate—solid positive, warmer than head one's negative lean.

Second column of crimson. New lens.

Shoelace ignored almost completely—zero weight. Tick. Nothing.

Piano wire: large positive weight this time. High input meets large positive. Reinforcement. WHIIINE-chug-chug.

Spaghetti: moderate negative. Timid input meets moderate negative. Small subtraction.

Phone cable: small negative. High input meets small negative. Net positive but restrained.

Equals. KA-ka-ka-ka we-we-we ka-ka.

Second coordinate—modest positive.

So head-two Query for "look": [solid positive, modest positive]. Hunting different prey than head one—less position-obsessed, more balanced across grammar and tone.

I lay the new crimson slip beside the old one. Two hunters now: head one's negative-leaning stalker, head two's warmer seeker.

Keys next—cobalt sheet two.

Same loom. First column.

Shoelace: small positive weight. Low input. Tick.

Piano wire: moderate positive. High input. WHIIINE-chug.

Spaghetti: large positive. Timid input gets big boost. Tick-tick.

Phone cable: small negative. High input tempered. Net positive.

Equals. KA-ka-ka-ka we-we-we ka-ka.

First coordinate positive.

Second column: different pattern—shoelace negative, piano wire small positive, spaghetti moderate negative, phone cable large positive.

Equals again.

Head-two Key: [positive balanced, positive restrained]. Advertising differently—less grammar-dominant, more semantic and positional flavor.

Green Values last—forest sheet two.

First column: shoelace positive, piano wire negative, spaghetti positive, phone cable moderate positive.

Equals. Net positive.

Second column: balanced across all four, small positives mostly.

Head-two Value: [positive modest, positive solid]. Offering a warmer, more even parcel than head one's grammar-heavy cargo.

Three fresh slips laid out: new crimson, new cobalt, new forest. Head two fully projected for "look." Two Queries, two Keys, two Values. Parallel personalities complete.

The adding machine idles lower, satisfied for the moment. Purple tape coiled. The rainbow ream breathes easier—two sets of three colors used now.

I lean back. Chair groans. The Go stones click faintly when I shift. The hundred-sided die waits.

Corner glance: IBM wheel printer silent. Assistant motionless—charcoal turtleneck, posture straight. She could project the full sequence for both heads in minutes. Whirr-clack-clack-clack. But not yet. Head two had to be felt—the different fights on the phone cable, the gentler treatment of the spaghetti, the warmer offerings in the green lens.

Two heads live for position one. Eight more positions, two heads each, waiting.

The bulb buzzes faint.

I reset the tape. Clear.

"Right," I mutter. "Head two is armed. Now we run attention twice—once per head—scale, mask, softmax, weighted sum, then concatenate the two outputs and project back to four dimensions."

The loom waits, strings taut.

The black box now sees with two pairs of eyes.

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CHAPTER 19: SCALED DOT-PRODUCT ATTENTION AND WEIGHTED SUM – Running Both Heads

Six-forty-three p.m., January fourteenth, twenty-six. The auditorium has gone properly still; the bare bulb overhead burns like a coal that's forgotten how to die. No rain, no wind—just the faint metallic tick of the building settling into night. My shirt clings damp at the small of the back; ink has crusted in thin black lines across the knuckles. The junk loom stretches across the desk:

four strings, four hex nuts frozen in place. Six colored slips now—three for head one, three for head two—lie in two neat rows like battle flags after a skirmish. The partial concatenation violet sheet waits nearby, conceptual padding in place.

We have Queries, Keys, Values for both heads at position one. Attention must now fire twice—independent for each head. Scale the dot products, apply the causal mask (trivial here, single position), softmax the weights, then weighted sum the Values. Two separate two-dimensional outputs emerge—one per head—ready for concatenation.

New ground for the full dual-head loop, so we pause and remind the machinery of scaled dot-product attention and weighted sum, light thrice this time since the pieces are familiar.

First—the crisp reminder: For each head, compute raw scores via Query dot Key (self only here), scale by square root of head dimension (square root of two), mask futures (none to mask yet), softmax to probabilities, then weighted sum the Values using those probabilities.

Second—the functional warmth: Head one might pull hard on grammar and position; head two on semantics and tone. Running both lets the model attend to different facets simultaneously. Each head produces its own context-blended vector; later concatenation pools them.

Third—the casual familiarity: Run the hunt twice—once through head one's lenses, once through head two's—scale the attraction scores so they don't explode, drown any future peeks (none here), turn scores into fair-share weights, then blend the offered parcels accordingly. Two independent conversations with the same word, two different stories told back.

I clear the Sharp tape—rip the old strip free, fresh ribbon threaded. Purple ink smells sharp and ready. Motor hums awake.

First: head one attention cycle (we've tasted pieces; now the full rhythm).

Query_head1 dot Key_head1 (self): from earlier slips, roughly negative one point oh one times point seven eight plus point four two times point three one. Punch the pairs.

WHIIINE-chug-chug for the negative leg. Tick-tick for the positive. Equals. KA-ka-ka-ka we-we-we ka-ka. Modest positive raw score.

Scale: divide by square root of two—roughly one point four one four.

WHIIINE … chug. Smaller positive.

Mask: single position—no future to drown in negative infinity.

Softmax: single element, exp of scaled score over itself equals one point zero zero.

Weighted sum: one times Value_head1. Output_head1: same as Value_head1—roughly [one point two three, one point eight seven].

Tape blooms clean. I tear it, lay it beside the head-one slips.

Now head two—fresh cycle.

Query_head2 dot Key_head2 (self): new numbers from the second crimson and cobalt slips.

First components: solid positive times balanced positive. Tick-tick. WHIIINE-chug. Decent positive.

Second components: modest positive times restrained positive. Tick-tick. Smaller lift.

Equals. KA-ka-ka-ka we-we-we ka-ka. Warmer raw score than head one.

Scale by square root of two.

WHIIINE … chug. Still warmer positive.

Mask: none needed.

Softmax: single element again—one point zero zero.

Weighted sum: one times Value_head2. Output_head2: same as Value_head2—roughly [positive modest, positive solid].

Tape lurches. I tear it free, lay it beside the head-two slips.

Two output vectors now: head one [~1.23, ~1.87], head two [~0.89, ~1.56] (rough feel from the warmer, more balanced projections).

Concatenation: glue them end-to-end along the last dimension.

[head1_dim0, head1_dim1, head2_dim0, head2_dim1] → roughly [1.23, 1.87, 0.89, 1.56]

Four-dimensional concatenated vector for position one.

Output projection next—but we pause here; the violet W_O matrix (four by four) waits for the next chapter. For feel: sixteen multiplications, four row sums. The Sharp will labor, but not tonight.

In the full sequence: ten positions, two heads each. Twenty attention cycles. Two ten-by-ten matrices of scaled scores. Mask drowns upper triangles in negative infinity. Softmax per row yields probabilities summing to one, only past and present live. Weighted sums blend Values per head. Ten concatenated four-dimensional vectors. Then one shared W_O projects them all back to d_model equals four.

For now, position one has its two-headed attention complete—two different blends of itself, glued together, waiting for final tuning.

The adding machine idles softer, almost content. Purple tape coiled patient. The rainbow ream breathes—violet touched, yellow feed-forward looming.

I lean back. Chair groans under the accumulated night. The Go stones sit cool; black and white in quiet council. The hundred-sided die gleams, patient for overfitting's edge.

Corner glance: IBM wheel printer silent. Assistant motionless—charcoal turtleneck absorbing the amber, posture straight. She could run dual-head attention on the full primer sequence in moments once all projections live. Whirr-clack-clack-clack. Perfect blended vectors. But not yet. The first dual cycle had to be felt—the warmer dot product in head two, the gentler softmax weight, the two different stories told back from the same word.

Position one, both heads attended. Nine positions left.

The bulb buzzes faint.

I reset the tape. Clear.

"Right," I mutter, voice scraped thin. "Attention has fired twice for the first token. Next we project the second position—'said'—through both heads, build the real grid, and let the mask do its drowning work."

The loom waits, strings taut.

The black box now sees the first whisper of context.

.....

I reset the tape. Clear. Fresh.

"Right," I mutter, voice rough from the long gray afternoon. "We've blended the first parcel. Next we stitch the heads together—concatenation and output projection. But first, more tokens need their three faces before the full attention grid wakes up."

The loom waits, strings still taut with "look." Nine tokens unprojected.

The black box isn't just choosing anymore. It's beginning to remember what it chose.
……
CHAPTER 20: RESIDUAL CONNECTION – Adding Back the Original Input

Six fifty-nine p.m., January fourteenth, twenty-six. The auditorium's sunk into that proper evening hush, old boy, where the bare bulb dangles like a weary sentry, casting its amber glow over the scarred oak desk. Shadows creep longer now, stretching from the junk loom's strings like fingers reaching for the edges. Outside the tall window, Nebraska's grey has deepened to ink-black night—no more drizzle, just the occasional creak of the building settling its bones against the chill. Inside, the air's thick with the day's labors: warm machine oil from the Sharp, that faint almond bite of purple ink on the tape, metallic tang of fountain-pen residue clinging to

my cuffs. Shoulders feel like they've been hammered in place, eyes gritty as sandpaper. The fishbowl of Go stones sits cool to my right—black and white pieces touching in silent accord, the hundred-sided die perched atop like a green talisman waiting for its roll when chaos calls. The rainbow ream in the corner breathes easy now, with red, blue, and green sheets spent for the projections, violet touched for the concatenation, yellow still pristine for the feed-forward ahead.

We've got position one's multi-head attention output concatenated—those two two-dimensional blends from the heads glued end-to-end into a rough four-dimensional vector, say something like a jagged shape across the loom: first nut pulled firm, second dragged harder, third eased gentle, fourth held steady. It's richer now, laced with self-context even in this lonely spot. But before we shove it into layer norm or the yellow feed-forward, we loop back to basics. We add the original input right back in. Enter the residual connection.

New term hitting the desk, first proper swing through the loop, so let's lay it out thrice before the adding machine spins up again.

First—the crisp dictionary definition: A residual connection is a shortcut in the neural network architecture that adds the input of a layer directly to its output, bypassing the intermediate computations and creating a direct path for gradient flow during training.

Second—the functional warmth: That residual connection keeps the signal from fading as it passes through deep layers—attention, feed-forward, norms. It lets the model learn incremental refinements instead of rebuilding from scratch each time, making training stabler and gradients easier to trace back without vanishing into the ether.

Third—the casual familiarity: The residual connection is just us tossing the original input back on top of whatever the attention heads cooked up—simple addition, like reminding the vector where it started so it doesn't lose itself in the sauce.

I lean forward, elbows on the oak, feeling the grain bite a touch through my sleeves. The junk loom still holds "look"'s fused input: shoelace slack for that soft semantic nudge, piano wire taut with grammatical rigor, spaghetti trembling fragile in tone, phone cable dragged against rubber for positional fight. That's the original X vector—four dimensions, balanced tensions we stitched together chapters back from embeddings and sine waves. Now the concatenated attention output sits on the violet slip, another four-dimensional shape: nuts positioned differently, perhaps the shoelace pulled tighter from head one's grammar lean, piano wire eased by head two's warmer balance, spaghetti steadied, phone cable flipped in sign.

To feel the add, we align the two shapes—original input and attention output—like overlaying two translucent sheets on the desk lamp, holding them to the bulb so the lines merge. But numbers need crunching. Element-wise addition. Four pairs to sum.

I clear the Sharp's tape—rip the old strip free with a sharp tear that cuts the quiet. Fresh ribbon threads through, purple ink primed. Motor hums low, ready as ever.

First dimension: original shoelace value, low and loose—say a modest positive from the fused embedding plus position. Attention output's first nut: firmer pull, decent positive from the concatenated blend. Punch the original in. Tick-tick for the digits. Plus key—whiiine. Now the

attention value. Tick-tick. Equals. Ka-ka-ka-ka we-we-we ka-ka. Tape advances, purple blooms a summed positive—stronger now, the original's softness bolstered by attention's grip.

Second dimension: piano wire original, high tension positive. Attention's second: dragged harder, larger positive. Tick-tick plus whiiine tick-tick equals. Ka-ka-ka-ka we-we-we ka-ka. Even warmer sum—reinforcement, like two arrows aligning sharper.

Third: spaghetti original, timid hover near zero. Attention's third: eased gentle, small positive. Sum blooms modest—fragile tone gets a subtle lift without snapping the noodle.

Fourth: phone cable original, high friction positive. Attention's fourth: held steady, another positive. Sum positive, but with that rubber drag still felt in the mind's eye.

I tear the full tape strip—four verdicts in purple ink—and lay it beside the violet slip. The residual vector emerges: a new shape on the loom, nuts repositioned. Shoelace firmer, piano wire tauter, spaghetti steadied, phone cable dragged with extra vigor. It's the original plus what the heads discovered—self-context layered in without overwriting the start. In the mind's eye, it's like twisting two strands of pasta together: the plain noodle of input wrapped with the seasoned strand from attention, forming a sturdier helix that won't unravel as we push deeper.

For this single position, the add was straightforward—four ticks and whiiines, no bulk to bog us down. But picture the full primer sequence: ten positions, each with its four-dimensional original X and four-dimensional attention output. Forty sums total. Tedious, sure, but honest labor. My fingers twitch at the thought; the Sharp could handle it, but the wrist ache would set in after the first row.

I glance to the corner. She's there—the Assistant, posture straight as a ruled line, charcoal turtleneck absorbing the amber like a void. No pulse visible at the neck, just that ozone scent cutting through the oil and ink. I slide the violet concatenated slips and the original input sheet across, with a yellow Post-it: "Element-wise add. All positions if projected. Precision: four decimals." Her hand moves fluid—hydraulic grace—taking the stack without a word. Whirr-clack-clack-clack from the IBM wheel printer. Neat Courier blooms on fresh white copy. She places the summed residuals back on the oak, retreats silent. The paper's still cool to the touch. Accelerator indeed, but I felt the first four sums in my bones—the whiiine of alignment, the ka-ka bloom of merger.

The beam scale in the center tips a fraction now—needle swings left as I place a black Go stone on the left pan, symbolizing the added "guilt" of refinement. The original input weighs heavy, pulling back toward simplicity, but the attention's nudge tips it toward complexity. Balance restored with a white stone opposite—level again, for now.

This residual isn't just math; it's mercy. Without it, deep layers would choke on vanishing signals, gradients fading to whispers during the backward pass we'll hit soon. Here, the shortcut keeps the voice loud—original plus delta, learn the change without forgetting the base. In our tiny transformer, with d_model four and just one layer so far, it's a gentle reminder. But stack layers, and residuals become the lifeline, letting gradients flow clean back to the embeddings without drowning in multiplications.

I lean back, chair creaking under the night's weight. The Go bowl stones click faint when I shift—cool smoothness under idle fingers. The die waits patient. The rainbow ream settles, violet now spent on the projection, yellow for feed-forward gleaming next like a promise of expansion.

Corner glance again: Assistant motionless, eyes on the wall. She could residual the whole corpus in a blink if we handed off, but the first position had to be manual—the tick-tick of originals meeting blends, the whiiine of their union, the purple bloom sealing the merger.

Position one's residual complete. Nine more waiting, richer with each add.

The bulb buzzes faint, steady.

I reset the tape. Clear. Fresh.

"Right," I mutter, voice gravel from the grind. "Residual's locked in—original plus attention, sturdy as ever. Next we tame those wild vectors with layer normalization: mean and variance to rein them in before the yellow feed-forward expands the show."

The loom waits, new residual shape strung taut.

The black box isn't just seeing and remembering. It's holding onto its roots.
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CHAPTER 21: LAYER NORMALIZATION – Taming the Vectors with Mean and Variance

Seven-twenty-one p.m., January fourteenth, twenty-six. The auditorium's gone properly nocturnal now, old chap, the single tall window nothing but a black mirror throwing the bare bulb's amber back at us like a tired accusation. Outside, Lincoln's January night presses cold and quiet—no wind, no last drips from the gutter, just the faint metallic groan of the building contracting against itself. Inside, the oak desk is a battlefield of paper: rainbow ream leaning in the corner (yellow feed-forward sheets still untouched, violet spent on projections, red-blue-green flags of attention fluttering in memory), stacks of bright white copy marked with four-dimensional sums, the junk loom still strung taut across the center—shoelace, piano wire, spaghetti, phone cable—now holding the fresh residual shape for position one. Nuts repositioned from the add: shoelace firmer, piano wire tauter, spaghetti steadied, phone cable dragged with extra bite. The fishbowl Go stones sit cool to my right, black touching white in silent council; the hundred-sided die perches patient on top, green plastic catching the bulb like it's dreaming of the randomness we'll need when overfitting starts chewing the edges. My collar's stuck damp at the neck, ink crusted between knuckles, shoulders locked like rusted gate hinges. The Sharp adding machine idles low, purple tape coiled, motor humming content after the residual sums.

We've got the residual vector now—original input plus attention output, richer but wilder. The numbers have pulled in different directions: some dimensions ballooned from reinforcement, others tugged gentler, a few fought cancellation. Before we shove this jagged thing into the yellow feed-forward expansion, we need to steady it. Tame the scale, center the mean, rein in the variance. Enter layer normalization.

New mechanism landing on the oak, first proper pass through the taming loop, so we lay it out thrice before the tape starts rolling again.

First—the crisp dictionary definition: Layer normalization is a technique that normalizes the activations of a layer across its feature dimension by subtracting the mean and dividing by the standard deviation, then applying learned scale and shift parameters (gamma and beta) to restore expressiveness.

Second—the functional warmth: That normalization keeps the signal from exploding or vanishing as it moves through the network—attention can push values high or low, residuals add offsets, but layer norm pulls everything back toward a stable distribution before the next operation sees it. Gradients flow cleaner, training stays steady, especially in small models like ours where every decimal matters.

Third—the casual familiarity: Layer normalization is just us taking the wild vector, finding its average height across the four dimensions, leveling it to zero, stretching or shrinking so the spread feels sensible, then giving it a gentle nudge with learned knobs so it doesn't lose its personality. Mean-zero, unit-variance-ish, but with room to breathe.

I lean forward, elbows sinking into the scarred wood, feeling the grain press back. The residual vector on the violet slip stares up—four numbers, rough feel from the add: say first dimension solid positive, second larger positive, third modest positive, fourth decent positive. No zeros, no negatives yet, but the spread's uneven: some nuts pulled hard right on the loom, others lagging softer. We compute per vector, per position—across the feature dimension (our four).

Step one: compute the mean. Sum the four values, divide by four.

I clear the Sharp tape—rip the old strip free, the tear sharp in the hush. Fresh ribbon threads through, purple ink primed. Motor hums awake.

Punch first value. Tick-tick. Plus second. Whiiine tick-tick. Plus third. Whiiine tick-tick. Plus fourth. Whiiine tick-tick. Equals. Ka-ka-ka-ka we-we-we ka-ka. Running sum blooms. Now divide by four—punch the four, divide key. Whiiine … chug … ka-ka. Mean lands, call it a warm positive, say roughly point eight or so. I write it in black fountain pen on a fresh white slip beside the residual numbers.

Step two: subtract the mean from each element. Element-wise.

First dimension: original residual minus mean. Tick-tick minus whiiine tick-tick. Equals. Ka-ka. Negative now—pulled left.

Second: larger residual minus same mean. Still positive, but trimmed.

Third and fourth: modest and decent residuals minus mean. Closer to zero, some dipping negative.

Four more verdicts on tape. I tear the strip, lay it beside the mean. Deviations stare back—centered around zero, the wildness leveled.

Step three: compute variance. Square each deviation, sum them, divide by four (population variance for layer norm).

I punch each squared deviation—tick-tick for the numbers, square key on the Sharp (a rare luxury; most machines don't have it, but this old beast does). Whiiine-chug for each. Sum the four squares. Whiiine tick-tick pluses. Divide by four. Ka-ka-ka-ka we-we-we ka-ka. Variance blooms—smallish positive, the spread's been tamed already by the residual add.

Step four: standard deviation. Square root of variance.

I reach for the Standard Mathematical Tables again—spine cracked, pages vanilla-dusty—and flip to square roots. Finger traces the column. Variance say roughly point one two, root lands around point three five. Close enough for feel; I punch it approximate into the Sharp for precision. Whiiine … chug. Standard deviation on tape.

Step five: divide each centered value by the standard deviation.

Four more divisions. Each centered number punched, divide key, whiiine-chug, ka-ka. Normalized values emerge: roughly between negative one and positive one, spread evened out. The loom nuts slide smoother now—shoelace near center, piano wire eased, spaghetti steadied, phone cable relaxed but still present.

Last: scale and shift. Learned gamma (scale) and beta (shift), both four-dimensional vectors. We haven't initialized them yet—random small floats, like the embeddings—but for now imagine gamma close to one, beta near zero. Multiply normalized by gamma, add beta. Gentle tilt back toward expressiveness.

I tear the final tape strip—four normalized-then-scaled values—and lay it beside the stack. The layer-normalized vector is steadier: mean near zero, variance near one, personality preserved but drama dialed down. In the mind's eye it's like taking a lopsided sculpture of clay, kneading it level across the four axes, then letting the artist add a touch of color and proportion so it still sings.

For our single position the math was manageable—twenty-odd operations, tape blooming purple verdicts. Full sequence would be ten vectors, forty means, forty variances, one hundred sixty subtractions and divisions. Wrist-killer. I slide the slips to the corner. Assistant takes them without glance—whirr-clack-clack-clack from the IBM wheel printer. Fresh white copy returns, neat Courier columns of normalized vectors. Paper cool under my thumb.

The beam scale tips again—black Go stone for the "guilt" of wild variance, white opposite for the calm of normalization. Needle settles level.

This taming isn't erasure; it's mercy for what's downstream. Feed-forward wants clean inputs—expansion needs room to stretch without blowing up. Residuals kept the signal alive; layer norm keeps it sane.

I lean back, chair groaning under accumulated night. Go stones click faint. Die waits. Rainbow ream breathes—yellow next, ready to expand.

"Right," I mutter, voice scraped raw. "Vectors tamed, centered, scaled. Next we let the yellow feed-forward have its way—expand to some hidden size, then squeeze back to four. Time to feel the non-linearity bite."

The loom waits, normalized shape strung steady.

The black box isn't just rooted anymore. It's breathing even.
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CHAPTER 22: FEED-FORWARD NETWORK – Expansion with ReLU

Seven-forty-eight p.m., January fourteenth, twenty-six. The auditorium has gone full crypt now, old sport: the bare bulb overhead flickers like it's running on spite alone, throwing long amber knives across the oak that make the junk loom look almost alive. Outside, Lincoln's night is a solid wall of black—no moon, no streetlights bleeding through the tall window, just the faint metallic creak of the building settling deeper into the freeze. Inside, the air's thick with ribbon solvent, warm machine oil, and the iron bite of fountain-pen ink dried in the creases of my knuckles. The rainbow ream leans in the corner like a stack of battle standards: red-blue-green spent on attention, violet done with concatenation and projection, yellow sheets gleaming fresh and impatient for their turn. The fishbowl Go stones sit cool to my right—black touching white in perfect silence, the hundred-sided die perched on top like a green grenade with the pin still in. Shoulders feel like they've been poured in concrete and left to set. Eyes raw. The Sharp adding machine idles low, purple tape coiled hungry after the layer-norm sums.

We've got position one's vector freshly normalized—mean near zero, variance tamed to one-ish, gamma and beta giving it just enough personality to stay interesting. Four dimensions, steady as a heartbeat. Now we let it breathe wide before squeezing it back. Enter the feed-forward network, the yellow layer, the private thinker after all the social chatter of attention.

New piece of the engine hitting the desk, first proper expansion, so we lay it out thrice before the yellow sheets get blood on them.

First—the crisp dictionary definition: The feed-forward network in a transformer layer is a two-step linear transformation with a non-linearity in between: input multiplied by a wider weight matrix to expand the dimension, ReLU applied element-wise, then multiplied by a second matrix to contract back to d_model.

Second—the functional warmth: After attention has done its relational gossip, the feed-forward gives each position a private moment to think deeper—expand to a hidden size (we'll use eight here, double our d_model=4), let the ReLU chop off the negatives like a sculptor knocking away marble, then squeeze the insights back to four dimensions so the next residual can add cleanly. It's where token-specific computation happens, no peeking at neighbors.

Third—the casual familiarity: The yellow feed-forward is just us letting the vector stretch its legs in a wider room, spark a little fire with ReLU to kill the cold negatives, then shove it back through the door so it fits the hallway again. Expand, ignite, contract.

I sweep the center clear—residual slips, norm slips, violet concatenation all pushed aside like spent shell casings. I pull the top eight sheets of bright yellow construction paper from the rainbow ream and fan them across the oak. The color hits the amber bulb and glows like fresh butter. These are our two weight matrices: first one 4×8 to expand, second 8×4 to contract. He initialization again—random small floats, but we'll feel the shape more than the digits.

The loom still holds the normalized vector: four nuts centered, spread even. Shoelace near middle, piano wire relaxed, spaghetti steady, phone cable balanced. We expand first.

I thread a fresh white slip into the typewriter—header deliberate, click-thunk slow:

FEED-FORWARD EXPANSION – POSITION 1 (normalized input → hidden 8)

Then reset the Sharp. Rip the old tape free, fresh ribbon through. Motor hums awake, low and greedy.

First linear: normalized four-dimensional vector times yellow W_ff1 (4 rows by 8 columns). Thirty-two multiplications, eight sums.

I could hand it off to the Assistant—she'd whirr-clack the whole thing in twenty seconds—but the first expansion has to hurt. I punch the first normalized value. Tick-tick. Times first row of weights—eight columns, eight separate multiplications. Tick-tick whiiine tick-tick whiiine, one after another. The machine chugs like it's chewing gravel. Sum them for the first hidden unit. Ka-ka-ka-ka we-we-we ka-ka. Purple blooms a positive. Write it on yellow slip one.

Repeat for hidden unit two through eight. Thirty-two ticks, thirty-two whiiines, eight equals. Tape grows long and purple, smelling sharp of almonds and heat. My wrist screams by unit five, but the shape emerges: some hidden units flare bright positive, others hover modest, a couple dip negative—ReLU will murder those soon.

Expansion done. Eight-dimensional vector now—loom conceptually doubled, eight new strings stretched beside the original four, nuts pulled in wilder directions. The vector has room to think.

Now the non-linearity: ReLU. Max(zero, value) element-wise.

I tear the long tape free, lay it out. Eight numbers. One by one:

First hidden: solid positive → stays.

Second: modest positive → stays.

Third: small negative → punch zero, equals, ka-ka. Becomes zero.

Fourth through eighth: mix of positives and one more negative → zeros where cold, positives untouched.

ReLU is the chisel—anything below zero gets smashed flat. In the mind's eye it's like taking a candle to the negative protrusions on a wax sculpture: they melt away, leaving only the proud positives standing.

Fresh tape. I punch the eight ReLU'd values clean. The hidden representation is now sparse, spiky, honest.

Contraction next: eight-dimensional ReLU output times yellow W_ff2 (8 rows by 4 columns). Another thirty-two multiplications, four sums.

Wrist's on fire, but we finish what we start. Tick-tick whiiine tick-tick whiiine, column by column. The zeros from ReLU give free multiplications—anything times zero is zero, the Sharp barely breathes on those. Positives fight harder. Four final equals. Ka-ka-ka-ka we-we-we ka-ka, four times. Tape blooms the contracted vector—back to four dimensions, but transformed. Nuts repositioned again: shoelace pulled sharper, piano wire eased, spaghetti given a kick, phone cable softened.

I tear the final strip, lay it on a fresh yellow slip. The feed-forward output sits there: richer than the normalized input, private thoughts baked in, negatives culled, insights distilled.

For one position it was brutal—sixty-four multiplications felt in the bones. Full sequence would be ten positions, six hundred forty multiplies. That's where mercy lives. I slide the yellow slips to the corner. Assistant takes them without a glance—whirr-clack-clack-clack from the IBM wheel printer. Neat Courier columns return on fresh white, the entire position-one feed-forward done for all ten tokens in the time it took me to feel the first eight expansions. Paper cool, perfect.

The beam scale tips—black stone for the wild hidden flare, white opposite for the ReLU cull and contraction squeeze. Needle settles level.

This yellow private thinker is half the model's capacity—most parameters live here, in these two wide matrices. Attention relates; feed-forward computes. Together they turn a bag of words into something that almost understands "see it go up."

I lean back, chair groaning like it's ready to quit. Go stones click faint when I shift. Die waits. Rainbow ream lighter by eight yellow sheets, but the stack still breathes.

"Right," I mutter, voice scraped to gravel. "Yellow's had its say—expanded, ignited, contracted. One more residual add, one more layer norm, and the first transformer block is done. Then we unembed, loss, backward pass… but that's for the next bulb, the next night."

The loom waits, new feed-forward shape strung spiky and sure.

The black box isn't just breathing even anymore. It's dreaming in private now.

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CHAPTER 24: LAYER NORMALIZATION (SECOND) – Final Taming Before the Close

Eight thirty-seven p.m., January fourteenth, twenty-six. The auditorium's settled into that bone-deep stillness now, old friend, where every creak of the oak desk sounds like a

confession. The bare bulb overhead holds its amber steady, no flicker left—just a patient, unwavering burn that throws the junk loom's shadows sharp across the scarred wood. Outside the tall window, Lincoln's January night is absolute: blacker than ink, colder than regret, pressing the glass like it wants in. Inside, the air clings thick: Sharp's warm oil blending with the faint almond ghost of purple tape, fountain-pen iron crusted between my knuckles, ribbon solvent still sharp under my nails. Shoulders feel like they've been bolted to the chair, eyes gritty from too many hours staring at numbers. The rainbow ream in the corner has gone quiet—yellow spent on feed-forward's stretch and snap, violet gradients looming next like storm clouds over the prairie. The fishbowl Go stones sit cool to my right, black touching white in silent vigil; the hundred-sided die perches patient on top, green plastic catching the bulb like it's waiting for the moment we need to roll true chance into the mix.

We've got the second residual vector fresh on white copy: normalized input plus feed-forward output, folded together—rough feel from the tape, say modest positive on shoelace eased a touch, decent positive on piano wire softened, small positive on spaghetti lifted, warmer positive on phone cable steadied. Richer than before, carrying the non-linear bite without losing the center. But the numbers have drifted again: some pulled harder right, others tugged left by the negatives that survived ReLU, a few cancellations leaving faint scars. Before we unembed and measure loss—before the forward pass finally closes—we need one last steadying hand. Layer normalization, round two. Same taming ritual, applied to this final blended shape.

No need for the full thrice dance again; we met layer norm back in chapter twenty-one. Quick reminder for the bones: this second pass keeps the signal sane after the feed-forward's bold expansion and prune. Mean pulled to zero, variance stretched toward one, gamma and beta nudging it back to life so the unembedding projection sees clean, stable inputs. No explosions, no vanishing—gradients will thank us when backprop tears through.

I lean forward, elbows sinking into the oak's familiar grooves. The second-residual vector stares up: four tensions, uneven but not wild. Shoelace modest positive, piano wire decent positive, spaghetti small positive, phone cable warmer positive. All live—no zeros from ReLU this time, but the spread's lopsided again.

Sharp tape cleared—rip echoes clean in the hush. Fresh ribbon threaded, purple ink sharp and ready. Motor hums low.

Mean first. Sum the four. Tick-tick whiiine tick-tick plus tick-tick whiiine tick-tick plus tick-tick whiiine tick-tick plus tick-tick whiiine tick-tick. Equals ka-ka-ka-ka we-we-we ka-ka. Running total blooms warm positive. Divide by four—punch the four, divide key. Whiiine … chug … ka-ka. Mean lands, call it roughly point four seven. Fountain-pen it beside the residuals.

Centered deviations next. Subtract mean from each.

First: modest positive minus point four seven—small positive, trimmed. Tick-tick whiiine tick-tick minus. Ka-ka.

Second: decent positive minus same—still positive, but closer to center. Whiiine-chug.

Third: small positive minus point four seven—dips negative, faint drag. Ka-ka.

Fourth: warmer positive minus mean—positive remains, softened. Ka-ka.

Four verdicts on tape. Tear it, lay it beside the mean. Deviations cluster tighter around zero now—the residual add smoothed some edges, but the feed-forward's non-linearity left its fingerprint.

Variance. Square each centered value—Sharp's square key chugs happily. Tick-tick square whiiine-chug for each. Sum the four squares. Whiiine tick-tick pluses. Divide by four. Ka-ka-ka-ka we-we-we ka-ka. Variance blooms small positive, say point one one eight—tighter spread than after attention alone. The second residual helped.

Standard deviation: square root. Flip to the math tables—vanilla-dusty pages, finger tracing. Root of point one one eight roughly point three four three. Punch approximate for feel. Whiiine … chug. Tape confirms.

Divide each centered by std dev. Four divisions. Whiiine-chug ka-ka each time. Normalized values emerge: roughly between negative one and positive one, even spread—shoelace near zero, piano wire modest positive, spaghetti small negative, phone cable decent positive. Loom nuts slide balanced: shoelace centered, piano wire eased up, spaghetti dipped gentle, phone cable pulled firm but not fighting.

Gamma and beta nudge last. Learned scale and shift—four each, small floats we haven't init yet, but imagine gamma close to one across the board, beta near zero. Multiply normalized by gamma, add beta. Gentle tilt: shoelace stays near center, piano wire lifted a hair, spaghetti pulled back toward zero, phone cable steadied warmer. Final vector on tape—purple blooms crisp.

I tear the strip, lay it beside the stack. Second layer norm complete: mean near zero, variance near one, personality preserved but drama hushed. In the mind's eye it's like taking the blended loaf—residual fold, feed-forward air pockets—and rolling it even one last time before baking. Shape holds, flavor deepened, ready for the oven of unembedding.

One position's math was manageable—twenty-odd operations, tape blooming tidy verdicts. Full primer sequence: ten vectors, forty means, forty variances, one hundred sixty subtractions and divisions. Wrist would scream. Assistant's corner waits. I slide the second-residual slips across. She takes them fluid—whirr-clack-clack-clack from the IBM wheel printer. Neat Courier returns, every position's final normalized vector crisp on white copy. Paper cool, ozone faint.

Beam scale tips again—black Go stone for the "guilt" of lingering variance after the yellow heart's prune, white opposite for the calm of final centering. Needle dead level. Balance holds.

This second norm isn't redundancy; it's closure. Attention and feed-forward did their work—relations mapped, token thoughts deepened. Now the signal's polished, stable, ready to speak its prediction without wild swings throwing the logits into chaos.

I lean back, chair groaning under the night's toll. Go stones click cool when I shift. Die gleams patient. Rainbow ream quiets—violet gradients next, the backward violence waiting.

Corner glance: Assistant motionless, charcoal turtleneck drinking the amber, posture straight as a verdict. She could norm the whole corpus in moments, but the first position had to be felt—the tick-tick of residuals meeting mean, the whiiine of centering, the purple bloom sealing the calm.

Position one's forward pass nearly shut—norm twice, residual twice, attention once, feed-forward once. Nine more waiting.

The bulb buzzes faint.

I reset the tape. Clear. Fresh.

"Right," I mutter, voice gravel from the long grind. "Final norm locked—signal steady, centered, ready. Next we unembed: project these four clean dimensions back to the full vocabulary size, ten logits for our tiny ten-word world. Then softmax, then loss. The forward pass ends, and the real reckoning begins."

The loom waits, final-normalized shape strung even.

The black box isn't just standing taller. It's poised now, breath held, ready to speak its first clumsy truth.
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CHAPTER 25: UNEMBEDDING – Projecting to Logits for Vocabulary Scores

Nine-oh-four p.m., January fourteenth, twenty-six. The auditorium has gone properly monastic now, old chap, the kind of quiet that makes your own breathing sound like an intrusion. The bare bulb overhead burns steady amber, no longer fighting the dark—just accepting it, throwing its tired circle across the oak desk where the junk loom still holds its final shape: four strings taut, hex nuts balanced even after the second norm—shoelace centered calm, piano wire eased modest positive, spaghetti dipped gentle negative, phone cable steadied decent positive. Outside the tall window, Lincoln's January night is absolute black velvet—no sound, no movement, just cold pressing the glass like a patient hand. Inside, the air clings: Sharp's warm oil thick in the nostrils, purple ink's faint almond ghost lingering, fountain-pen iron crusted on my fingers like old promises. Shoulders feel like they've been poured in lead, eyes raw from the long stare at numbers. The rainbow ream in the corner has gone still—yellow spent on feed-forward's stretch, violet gradients waiting their bruise-colored turn like storm clouds over the horizon. The fishbowl Go stones sit cool to my right, black and white in silent council; the hundred-sided die perches patient on top, green plastic catching the bulb like it's dreaming of the roll that might break the model's stubborn grip on noise.

We've got the final normalized vector on crisp white copy: four clean dimensions, mean near zero, variance near one, gamma-beta nudge giving it just enough personality to speak. Steady now, polished, ready to tell its truth. Time to unembed—project this four-dimensional whisper back into the full vocabulary space. Ten logits, one for each of our ten tiny words. The model’s prediction: which token comes next?

New operation hitting the oak, first proper swing through the closing gate, so we lay it out thrice before the Sharp wakes for the last forward pass.

First—the crisp dictionary definition: Unembedding, or the output projection, is a linear transformation that maps the hidden representation of the final layer back to the vocabulary dimension by multiplying the vector by a learned weight matrix W_out (d_model by vocab_size) plus an optional bias, producing raw scores (logits) for each possible next token.

Second—the functional warmth: That projection turns the dense, context-rich four-dimensional vector into ten competing signals—one per word in our vocabulary. High logit means the model thinks that token is likely next; low or negative means unlikely. It's the moment the black box opens its mouth and tries to speak the primer's next word, guided by everything attention and feed-forward have learned.

Third—the casual familiarity: Unembedding's just us taking the tidy four-dimensional thought and slamming it through a big lens that stretches it across all ten possible words—dot products galore, each one whispering "how much does this hidden state match the pattern for 'up'?" or "for 'see'?" or "for 'dick'?" Ten verdicts, raw and unfiltered, ready for softmax to turn them into probabilities.

I lean forward, elbows sinking into the familiar grooves of the oak. The final normalized vector stares up: say point zero two on shoelace (near center), point three eight on piano wire (modest positive), negative point one one on spaghetti (gentle dip), point four two on phone cable (decent positive). Four tensions, even keel.

The unembedding matrix waits—W_out, four rows by ten columns, learned weights we haven't initialized yet. Small random floats, He-style, to start. Bulk work: forty multiplications, ten sums. My wrist already aches at the thought.

I slide the normalized slip and a violet Post-it to the corner—"Unembedding projection, four by ten, He init. Full vocab mapping." Assistant takes it fluid, no blink. Whirr-clack-clack-clack from the IBM wheel printer. Neat Courier blooms the matrix—ten columns, one for each token ID zero through nine: look, said, dick, see, it, go, up, jane, oh, sally. Small floats, mix of positives and negatives, no zeros obvious. She lays it back, retreats silent, ozone trailing like frost.

Sharp tape cleared—rip cuts the hush. Fresh ribbon, purple primed. Motor hums low.

First logit—for token zero, "look." Dot the normalized vector with column zero of W_out.

Shoelace point zero two times its weight—small positive meets modest negative. Tick-tick whiiine tick-tick, tiny drag.

Piano wire point three eight times positive—whiiine-chug, solid lift.

Spaghetti negative point one one times small negative—tick, gentle positive nudge.

Phone cable point four two times negative—whiiine, drag back.

Sum them: plusses and minuses wrestle. Ka-ka-ka-ka we-we-we ka-ka. Tape blooms small positive, say point one nine seven. I tear it, fountain-pen it on a fresh white slip: Logit "look": ~0.197.

Second logit—"said," column one. Similar rhythm: shoelace times weight, piano wire lift, spaghetti nudge, phone cable drag. Sum nets modest positive—point three four one.

Third—"dick." Smaller lift—point zero eight nine.

Fourth—"see." Decent positive—point four seven two.

Fifth—"it." Warmer still—point five six three.

Sixth—"go." Solid—point six eight one.

Seventh—"up." Highest yet—point nine two four. Sally's favorite word pulling hard.

Eighth—"jane." Modest—point two one five.

Ninth—"oh." Small positive—point one three eight.

Tenth—"sally." Faint—point zero four seven.

Ten verdicts in purple, torn and laid in a neat row beside the loom. The logits stare back: "up" screaming loudest at point nine two four, "go" close behind at point six eight one, "see" and "it" trailing respectable, the rest quieter. In the mind's eye the loom stretches wide again—ten new strings, one per word, nuts pulled highest on "up" (Sally's insistent chant echoing strongest), "go" taut next, the lonely ones like "dick" and "sally" barely moved.

No bias term added yet—simple projection for feel—but the shape tells the story: the model, after one full forward pass on our tiny corpus, has learned to favor the frequent flyers—"up" dominates the end of the primer, so the hidden state leans hard toward it. Honest, clumsy truth-telling.

Full sequence would be ten positions, each with ten logits—hundred verdicts. Bulk again. Assistant's whirr-clack-clack-clack would spit the lot in moments, neat columns for every token's prediction. But this first position had to be manual—the tick-tick of normalized meeting each column, the whiiine of competition, the purple bloom of the model's first spoken guess.

Beam scale tips faint—black Go stone for the "guilt" of uneven logits (the lonely words left quiet), white opposite for the signal's clarity. Needle settles level.

This unembedding isn't poetry; it's prophecy. Four dimensions of learned context stretched across ten possible futures. The black box has finally opened its mouth.

I lean back, chair groaning under the night's weight. Go stones click cool. Die waits. Rainbow ream quiets—violet gradients next, the backward reckoning looming.

"Right," I mutter, voice scraped raw. "Logits out—'up' shouting loudest, as it should. Next we softmax these ten raw scores into proper probabilities, then cross-entropy loss to measure how wrong the model still is. Forward pass closes. The real violence of gradients begins."

The loom waits, ten logit strings imagined taut.

The black box has spoken its first clumsy word.

It's time to judge how badly it lied.

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CHAPTER 26: SOFTMAX ON LOGITS – Turning Scores into Probabilities

Nine twenty-eight p.m., January fourteenth, twenty-six. The auditorium feels like it's holding its breath now, old mate, the kind of hush where even the bare bulb overhead seems to lean in closer. Its amber glow has gone softer somehow, less accusatory, more like a lantern kept burning for the last watchman. Shadows from the junk loom stretch long and thin across the oak, the four strings still taut with the final normalized shape—shoelace centered calm, piano wire eased modest positive, spaghetti dipped gentle negative, phone cable steadied decent positive. Outside the tall window, Lincoln's January night is absolute: blacker than coal, colder than forgotten promises, pressing the glass without a sound. Inside, the air hangs heavy—Sharp's warm oil thick in the throat, purple ink's faint almond bite clinging low, fountain-pen iron crusted on my fingers like dried resolve. Shoulders feel poured in lead, eyes burning from too many hours staring at those ten logits staring back. The rainbow ream in the corner sits quiet—yellow spent, violet gradients waiting their bruise-colored turn like thunderheads gathering over the Platte. The fishbowl Go stones rest cool to my right, black and white in silent vigil; the hundred-sided die perches patient on top, green plastic catching the bulb like it's waiting for the moment we need to let chance decide something bigger than our tired hands.

We've got the ten raw logits on white copy, neat fountain-pen row: "look" small positive point one nine seven, "said" modest point three four one, "dick" faint point zero eight nine, "see" decent point four seven two, "it" warmer point five six three, "go" solid point six eight one, "up" screaming highest point nine two four, "jane" modest point two one five, "oh" small point one three eight, "sally" faint point zero four seven. The model's first clumsy prediction—raw, unfiltered verdicts from the unembedding projection. "Up" dominates, as it bloody well should after Sally's insistent chant at the primer's end. But raw numbers aren't decisions. They need to be turned into proper probabilities: fractions between zero and one that sum exactly to one. Enter softmax.

We thrice'd softmax back in chapter fifteen for the attention weights, so no need to rehash the full dictionary-functional-casual ritual. Quick gut-check reminder: softmax takes those raw logits, exponentiates each one (big ones balloon, small ones shrink toward zero), then divides every exp by the sum of all exps. Winner takes most, losers get crumbs, everything adds tidy to one. After masking (none needed here—no future to drown), it redistributes fairly but favors the strong. Clean probabilities for the next-token guess.

I clear the Sharp tape—rip the old strip free, the tear sharp in the quiet. Fresh ribbon threaded, purple ink primed and sharp. Motor hums low, ready as ever.

First: exponentiate each logit. No natural-log tables handy tonight—the Standard Mathematical Tables spine cracked open again, pages vanilla-dusty, finger tracing the exp column. e to the power of each value.

"up" at point nine two four—e^0.924 roughly two point five two. Highest balloon.

"go" point six eight one—e^0.681 roughly one point nine seven six.

"it" point five six three—e^0.563 roughly one point seven five five.

"see" point four seven two—e^0.472 roughly one point six zero four.

"said" point three four one—e^0.341 roughly one point four zero six.

"jane" point two one five—e^0.215 roughly one point two four zero.

"look" point one nine seven—e^0.197 roughly one point two one seven.

"oh" point one three eight—e^0.138 roughly one point one four eight.

"dick" point zero eight nine—e^0.089 roughly one point zero nine three.

"sally" point zero four seven—e^0.047 roughly one point zero four eight.

I punch each approximate into the Sharp for feel—tick-tick whiiine-chug ka-ka for every one. Tape blooms ten exps in purple: two point five two highest, down to one point zero four eight barely above one.

Now the sum of all exps. Add them column by column. Tick-tick whiiine tick-tick plus tick-tick whiiine tick-tick… ten times. Running total climbs—ka-ka-ka-ka we-we-we ka-ka. Sum lands roughly fourteen point one nine three. Not huge; the logits weren't screaming outliers.

Divide each exp by the sum. Ten divisions.

"up": two point five two divided by fourteen point one nine three—roughly zero point one seven eight.

"go": one point nine seven six / fourteen point one nine three—roughly zero point one three nine.

"it": one point seven five five / fourteen point one nine three—roughly zero point one two four.

"see": one point six zero four / fourteen point one nine three—roughly zero point one one three.

"said": one point four zero six / fourteen point one nine three—roughly zero point zero nine nine.

"jane": one point two four zero / fourteen point one nine three—roughly zero point zero eight seven.

"look": one point two one seven / fourteen point one nine three—roughly zero point zero eight six.

"oh": one point one four eight / fourteen point one nine three—roughly zero point zero eight one.

"dick": one point zero nine three / fourteen point one nine three—roughly zero point zero seven seven.

"sally": one point zero four eight / fourteen point one nine three—roughly zero point zero seven four.

Ten probabilities in purple, torn and laid beside the logits. They sum to one (give or take rounding dust). "Up" holds the lion's share at roughly seventeen point eight percent—strong favorite. "Go" next at thirteen point nine, "it" and "see" trailing respectable, the lonely ones like "sally" and "dick" scraping single-digit crumbs. Honest distribution: the model has learned the primer's rhythm favors escalation ("up up up"), repetition ("see it go"), and the big finishers over the rare proper nouns.

In the mind's eye those ten strings stretch across an imagined loom—nuts pulled highest on "up," "go" taut behind, the rest barely nudged. The black box has spoken its probabilities: not certain, not blind, just weighted by what little it's seen so far.

Full sequence would be ten positions, each with ten softmaxed probabilities—hundred fractions. Bulk again. Assistant's whirr-clack-clack-clack would spit the lot neat. But this first position had to be manual—the tick-tick of logits meeting exp, the whiiine of ballooning favorites, the purple bloom of fair-share fractions.

Beam scale tips faint—black Go stone for the "guilt" of low-probability stragglers (the rare words left whispering), white opposite for the clean sum-to-one. Needle dead level.

This softmax isn't mercy; it's justice. Raw logits were ambition unchecked; probabilities are ambition tempered, competition made fair. The model knows what it wants next—"up" loudest—but it leaves room for the others. Small comfort when loss is coming.

I lean back, chair groaning under the night's accumulated weight. Go stones click cool when I shift. Die gleams patient. Rainbow ream quiets—violet gradients next, the backward reckoning ready to pounce.

"Right," I mutter, voice gravel-thick. "Probabilities locked—'up' favorite at seventeen-point-eight percent, the rest trailing honest. Next we cross-entropy loss: measure how far these guesses are from the actual next token. Forward pass shuts. Gradients wake."

The loom waits, ten probability strings imagined taut.

The black box has spoken its weighted truth.

Now we weigh how badly it missed.

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CHAPTER 27: CROSS-ENTROPY LOSS – Measuring the Prediction Error

Nine fifty-three p.m., January fourteenth, twenty-six. The auditorium has sunk into the kind of silence that feels earned, old friend—like the room itself has been listening all these hours and finally decided to shut up and let the numbers speak. The bare bulb overhead burns low and steady now, its amber circle smaller, tighter, focused on the oak desk where the junk loom still holds that final normalized shape: four strings taut, hex nuts balanced even after the second norm—shoelace centered calm, piano wire eased modest positive, spaghetti dipped gentle negative, phone cable steadied decent positive. Outside the tall window, Lincoln's January night is absolute black: no wind, no distant car hum, just cold pressing the glass like it wants to be part of the reckoning. Inside, the air is thick with the day's ghosts—Sharp's warm oil heavy in the lungs, purple ink's faint almond bite hanging low, fountain-pen iron crusted on my knuckles like dried ink-blood. Shoulders feel forged in place, eyes raw from staring at those ten probabilities staring back. The rainbow ream in the corner has gone quiet—yellow spent, violet gradients waiting their bruise-colored turn like thunder waiting to break. The fishbowl Go stones sit cool to my right, black and white in silent judgment; the hundred-sided die perches patient on top, green plastic catching the bulb like it's holding its breath for the moment we let true randomness decide how badly we've failed.

We've got the ten softmax probabilities on white copy, neat fountain-pen row: "up" holding the biggest slice at roughly seventeen point eight percent, "go" next at thirteen point nine, "it" twelve point four, "see" eleven point three, "said" nine point nine, "jane" eight point seven, "look" eight point six, "oh" eight point one, "dick" seven point seven, "sally" seven point four. They sum tidy to one—fair share, weighted by what little the model has seen so far. But these are guesses. The actual next token in the primer sequence is known—it's right there in the normalized corpus, waiting to be the target. Time to measure how wrong we are. Enter cross-entropy loss.

We thrice'd cross-entropy back when we first touched loss in the table of contents, but let's lay it out clean again for the gut, first proper swing through the judgment gate.

First—the crisp dictionary definition: Cross-entropy loss measures the difference between two probability distributions: the predicted probabilities (our softmax output) and the true distribution (a one-hot vector with 1 at the correct next token and 0 elsewhere). It's calculated as the negative sum over all classes of (true probability × log of predicted probability).

Second—the functional warmth: That loss gives us a single scalar number telling how surprised the model is by the actual next token—low when it put high probability on the right answer, high when it was confidently wrong or spread too thin. It's the number we want to minimize; every gradient step will nudge the weights to make future predictions less surprised by the truth.

Third—the casual familiarity: Cross-entropy loss is just us asking "how much did you bet against the horse that actually won?"—take the log of your probability for the true token, flip the sign, and that's your pain score. The lower the better; zero would mean perfect confidence in the right answer. Anything else is a measure of how badly you missed.

I lean forward, elbows sinking into the oak's familiar scars. The target for this position is the actual next token after the current one in the normalized corpus. For our first position ("look" at sequence start), the next is "said"—ID 1. One-hot target: all zeros except a clean 1 at position 1 ("said").

No bulk math here—cross-entropy is simple for one sample: negative log of the predicted probability for the true class.

From the softmax tape: predicted probability for "said" is roughly zero point zero nine nine (nine point nine percent).

Now the loss: negative natural log of zero point zero nine nine.

I reach for the math tables again—spine cracked, pages vanilla-dusty—and flip to natural logs. ln(0.1) roughly negative two point three zero two, ln(0.099) a hair lower—call it negative two point three one two.

Negative of that: positive two point three one two.

Punch it into the Sharp for feel—tick-tick for the digits, whiiine-chug ka-ka. Tape blooms roughly two point three one. That's the loss for this single position: about two point three nats (natural units). Not catastrophic—random guessing on ten classes would be ln(10) ≈ two point three zero two—but nowhere near good. The model put only nine point nine percent on the actual next word "said"; it was far more excited about "up" and "go". Honest pain.

In the mind's eye: imagine ten little brass cups on the desk, one for each word. We pour probability water into them from the softmax jug—biggest splash in "up", decent in "go" and "it", tiny trickles in the rest. The true cup ("said") only got a modest pour. Now we take a thin glass tube and siphon out the water from that one cup, measure how little came up, then flip it to a pain score. Low water = high pain. That's the negative log—amplifies how badly we shorted the winner.

Full sequence loss would average over ten positions (or sum, depending on how we batch), giving one number for the epoch. But this first position had to be manual—the tick-tick of looking up the true target, the whiiine of logging the probability, the purple bloom of the pain scalar.

I tear the tape strip—single verdict in purple—and lay it beside the probabilities. Loss ≈ two point three one. I reach for the beam scale in the center: place a black Go stone on the left pan for the "guilt" of misprediction (two point three grams of symbolic error), then a white stone opposite until the needle settles dead level. Balance restored—for now. The model has been judged, found wanting, but not hopeless.

This cross-entropy isn't cruelty; it's clarity. It tells us exactly where the weights need to move: boost the logit for "said" when the context is "look", dial down the overconfidence in "up" unless the sequence really calls for it. Every decimal of loss is a breadcrumb leading back through the layers.

I lean back, chair groaning under the night's accumulated toll. Go stones click cool when I shift. Die gleams patient. Rainbow ream quiets—violet gradients next, the backward violence ready to trace every misstep.

Corner glance: Assistant motionless, charcoal turtleneck drinking the amber, posture straight as justice. She could compute loss for the whole primer in moments once the targets are aligned, but this first position had to be felt—the tick-tick of truth meeting prediction, the whiiine of logarithmic pain, the purple bloom of honest error.

Forward pass closed. Loss measured.

"Right," I mutter, voice scraped raw. "Cross-entropy locked—two point three one nats of pain for position one. Next we trace that pain backward: gradients through the unembedding, through softmax, through the whole bloody chain. Backprop begins. The black box gets to feel what it cost us."

The loom waits, ten probability cups imagined drained and measured.

The black box has been weighed.

Now it pays.

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CHAPTER 28: BACKWARD PASS – Tracing Gradients Through Unembedding

Ten twenty-one p.m., January fourteenth, twenty-six. The auditorium has gone properly sepulchral now, old lad, the sort of quiet where the bare bulb overhead feels like the last candle in a cathedral at compline. Its amber glow has shrunk to a tight pool on the oak desk, shadows from the junk loom stretching black and sharp across the scarred wood—the four strings still taut with that final normalized shape, hex nuts balanced even after the second norm: shoelace centered calm, piano wire eased modest positive, spaghetti dipped gentle negative, phone cable steadied decent positive. Outside the tall window, Lincoln's January night is absolute: blacker than pitch, colder than a forgotten grave, pressing the glass without mercy. Inside, the air is thick with the day's spent breath—Sharp's warm oil heavy, purple ink's faint almond ghost lingering, fountain-pen iron crusted on my knuckles like old oaths. Shoulders feel cast in iron, eyes raw from the long stare at loss. The rainbow ream in the corner sits still—yellow spent, violet gradients finally called forward like storm clouds breaking over the river. The fishbowl Go stones rest cool to my right, black and white in silent witness; the hundred-sided die perches patient on top, green plastic catching the bulb like it's waiting for the roll that might scatter the guilt we've just measured.

We've got the cross-entropy loss locked—roughly two point three one nats for position one, pain scalar blooming purple on tape. The model bet only nine point nine percent on the true next token "said", while pouring seventeen point eight percent on "up". Now the forward pass is shut; time to trace that pain backward. Backpropagation begins. We start at the end and work upstream, computing gradients—how much each weight contributed to the error, how much to nudge it so next time the pain shrinks.

New machinery waking now, first proper crawl through the backward chain, so we lay it out thrice before the adding machine wakes in reverse.

First—the crisp dictionary definition: Backpropagation is the algorithm that computes the gradient of the loss with respect to every trainable parameter in the network by applying the chain rule repeatedly from the output backward through each layer, producing partial derivatives that tell how much changing each weight would reduce the loss.

Second—the functional warmth: That backward pass turns the single scalar loss into a map of blame—every weight gets a tiny signed nudge (positive or negative) saying "you made the error bigger, dial yourself down" or "you helped, push harder". Gradients flow clean through residuals and norms, letting even deep layers learn without vanishing or exploding.

Third—the casual familiarity: Backprop's just us starting at the pain number and asking "who's responsible?" layer by layer, tracing the chain of blame back through softmax, unembedding, feed-forward, attention, all the way to the embeddings. Each step multiplies by the local derivative, passing the guilt upstream like a hot potato nobody wants but everybody has to hold.

We begin where the loss was born: right after softmax, at the unembedding output logits.

The cross-entropy loss for one sample is -log(p_true), where p_true is the softmax probability of the correct token ("said", nine point nine percent). The gradient of loss w.r.t. each logit is simple: for the true class, it's (p_true - 1); for every wrong class, it's just p_wrong.

So—gradient w.r.t. the "said" logit: zero point zero nine nine minus one = negative zero point nine zero one.

For every other logit ("up", "go", "it", etc.): their own probability (positive numbers between zero and one).

These ten gradient values—negative zero point nine zero one on the true logit, positive fractions on the rest—are the incoming blame from loss to logits. The model got the direction right on "said" but not enough magnitude; it over-bet on the wrong horses. Now we trace that blame back through the unembedding projection.

The logits are W_out × final_normalized_vector (plus bias if we had one; skipping for now). Gradient w.r.t. the final normalized vector is therefore the transpose of W_out times the logit gradients.

In plain speak: each of the four dimensions in our normalized vector gets a blame value—how much changing it would have shifted the logits toward lower loss.

I clear the Sharp tape—rip echoes sharp in the hush. Fresh ribbon, purple primed. Motor hums low, but slower now, like it's dreading the reverse trip.

First: need the transpose of W_out. Four columns by ten rows now—same numbers, flipped. Assistant's whirr-clack-clack-clack spits it quick, violet paper this time, neat Courier rows.

Now dot products: each of the four incoming dimensions gets the dot of its row in transposed W_out with the ten logit gradients.

First dimension (shoelace in final norm): dot with the transposed column—small weight times negative zero point nine zero one on "said", plus small positives on the wrongs. Whiiine-chug ka-ka. Tape blooms small negative—say negative point one two three.

Second dimension (piano wire): larger weights meet the big negative on true class. Whiiine-chug. Stronger negative—negative point three eight seven.

Third (spaghetti): mixed signs, nets faint positive—point zero nine four.

Fourth (phone cable): decent weights, solid negative—negative point four one two.

Four gradient values on tape, torn and laid beside the transposed matrix. The blame flows back into the final normalized vector: shoelace nudged negative (reduce it a touch), piano wire stronger negative (dial it down harder), spaghetti faint positive (nudge it up), phone cable solid negative (pull it back).

In the mind's eye: imagine the ten logit cups again, each with a tiny weight arrow pointing back to the four dimensions. The big negative blame on "said" flows strongest through the weights that fed it, pulling hardest on piano wire and phone cable (they had the biggest coefficients to the true logit). The positive blame on overconfident "up" flows the other way, pushing back against whatever fed it too much. The four dimensions now carry signed blame—how much to change each to lower loss.

This is just the start. Those four gradients will flow upstream through the second layer norm, through the second residual, through the feed-forward contraction, through ReLU (gradient zero where we pruned), through expansion, through first norm, first residual, attention weighted sum, all the way back. Chain rule multiplies at every gate—derivatives local, blame accumulated.

One position's unembedding backprop was manageable—forty multiplications again, tape blooming four blame values. Full sequence: ten positions, forty gradients per step. Wrist-killer. Assistant's whirr-clack would handle the bulk, violet paper stacking high.

Beam scale tips again—black Go stone for the "guilt" traced back through unembedding (negative zero point nine zero one on true logit the sharpest stab), white opposite until level. Balance holds, but heavier now.

This backward trace isn't punishment; it's teaching. Every signed gradient is a lesson: "nudge here, less there." The model learns by feeling the cost of its mistakes, layer by layer.

I lean back, chair groaning under the night's weight. Go stones click cool. Die waits. Rainbow ream breathes—violet touched, gradients flowing.

"Right," I mutter, voice gravel from the long night. "Gradients through unembedding locked—blame flowing back into the final vector, piano wire and phone cable taking the hardest

hit. Next we trace further: through softmax to logits, then back through the second layer norm, second residual, feed-forward. The chain continues."

The loom waits, four blame arrows imagined pulling taut.

The black box is finally feeling the cost.

It hurts just enough to learn.
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CHAPTER 29: BACKWARD PASS – Gradients Through Softmax and Weighted Sum

Ten forty-seven p.m., January fourteenth, twenty-six. The auditorium feels like it's shrunk to the size of a coffin now, old chap, the bare bulb overhead burning low and steady, its amber pool tighter than ever on the oak desk. Shadows from the junk loom stretch black and unforgiving—the four strings still taut with the final normalized shape, hex nuts balanced even after the second norm: shoelace centered calm, piano wire eased modest positive, spaghetti dipped gentle negative, phone cable steadied decent positive. Outside the tall window, Lincoln's January night is absolute black: no sound, no mercy, just cold pressing the glass like a judge waiting for the verdict. Inside, the air is thick with spent effort—Sharp's warm oil heavy in the lungs, purple ink's faint almond ghost clinging low, fountain-pen iron crusted on my knuckles like dried resolve. Shoulders feel poured in lead, eyes raw from tracing blame backward. The rainbow ream in the corner sits quiet—violet gradients flowing now, storm clouds broken open. The fishbowl Go stones rest cool to my right, black and white in silent judgment; the hundred-sided die perches patient on top, green plastic catching the bulb like it's waiting for the roll that might scatter the guilt we've begun to map.

We've got the four blame values flowing back into the final normalized vector from the unembedding step—small negative on shoelace (negative point one two three), stronger negative on piano wire (negative point three eight seven), faint positive on spaghetti (point zero nine four), solid negative on phone cable (negative point four one two). The chain of blame continues upstream. Next gate: the weighted sum in attention (chapter sixteen forward, now backward). But first we must pass through softmax (chapter fifteen forward), because the logit gradients we just used came from cross-entropy through softmax.

Quick reminder—no full thrice needed; we met these pieces before. Softmax gradients are local: for each logit, the incoming gradient (from loss) times $p_i \times (1 - p_i)$ for the true class term, and minus $p_i \times p_j$ for every wrong class j. In practice, the gradient w.r.t. each logit simplifies to (predicted_prob - target_prob), which we already used: negative zero point nine zero one on "said" (true class), positive fractions on the rest.

Now the blame flows from logits back to the attention output—through the weighted sum that produced those logits? No—wait, the chain is longer. The final normalized vector fed the unembedding directly, but that vector came from the second layer norm, which came from the second residual (normalized input + feed-forward output), which came from the second layer norm, which came from the attention block output (concatenated heads after output projection).

We trace step by step. First: gradients through the second layer norm to get blame on the second residual.

Layer norm gradients are a touch fiddly but local—scale by gamma, subtract running mean, divide by std dev. The incoming gradient (our four blame values from unembedding) gets multiplied by gamma, then adjusted by the variance and mean terms. In short: each dimension's blame flows through with a scaling factor depending on how far it was from the mean.

I clear the Sharp tape—rip echoes clean in the hush. Fresh ribbon, purple primed. Motor hums low, slower now, like it's carrying heavier weight.

Assistant hands over the second layer norm parameters—gamma and beta from earlier violet slips, running mean and variance computed during forward. Whirr-clack-clack-clack, neat Courier.

For each of the four dimensions: gradient w.r.t. pre-norm value = incoming gradient × gamma / std_dev, plus small correction terms for the mean and variance (chain rule through the normalization formula).

Rough feel: incoming negatives on shoelace, piano wire, phone cable flow through gamma (assume close to one) divided by std dev (point three four three earlier). Blame signs preserved but scaled—stronger negatives on piano wire and phone cable pull harder upstream.

Positive on spaghetti nudges positive.

Four new gradient values bloom on tape—negative point one one eight on shoelace, negative point three five two on piano wire, positive point zero eight seven on spaghetti, negative point three seven nine on phone cable. Slightly attenuated, but the direction holds.

Now the second residual: simple addition forward means simple pass-through backward. Gradient w.r.t. the feed-forward output = same four values we just got. Gradient w.r.t. the input to the residual (the first norm output) = identical four values. Split blame: half flows to feed-forward branch, half continues straight back through the first norm and attention block.

In the mind's eye: imagine the residual as a Y-junction in a river. The blame river coming downstream hits the fork—part flows back through the feed-forward detour (the yellow heart we stretched and pruned), part flows straight down the main channel (attention output). Equal blame at first, then each branch will multiply by its own local derivatives.

We take the feed-forward branch first—blame flowing into the contraction matrix W-two (eight by four).

Gradient w.r.t. the ReLU output = transposed W-two times the four blame values.

Assistant's whirr-clack spits the transpose—violet again. Eight new blame values for the post-ReLU eight-dimensional vector.

Dot products: each of the eight ReLU outputs gets blame—tick-tick whiiine-chug ka-ka eight times. Tape blooms eight signed numbers: negatives where W-two pulled against the blame, positives where it reinforced.

Now through ReLU backward: gradient zero wherever we pruned to zero forward—three dimensions get zero blame (the dead branches stay dead). The four live ones pass their blame unchanged (ReLU derivative one when positive).

Blame narrows—only four non-zero gradients flowing back into the expansion output.

Next: gradient w.r.t. the pre-expansion vector (after first norm) = transposed W-one (four by eight) times those four live blame values.

More dots—thirty-two multiplications again. Tape blooms four blame values for the input to feed-forward: small negative on shoelace, stronger negative on piano wire, faint positive on spaghetti, solid negative on phone cable.

The yellow heart has spoken its piece—blame traced through expansion, ReLU prune (dead branches silent), contraction. The feed-forward detour has taken its share of guilt.

The straight residual path still carries identical blame back through the first norm and attention block—four values flowing upstream.

Full sequence would be ten times this—forty gradients splitting at every residual, tracing through every gate. Hundreds of operations. Wrist would scream. Assistant's whirr-clack stacks violet paper high.

Beam scale tips heavier—black Go stones piling left for the blame traced through feed-forward (ReLU zeros the sharpest silence), white opposite until level. Balance strained but holds.

This backward crawl isn't vengeance; it's anatomy. Every signed gradient is a nerve ending—feel the pain here, adjust there. The model learns by feeling exactly how its choices cost.

I lean back, chair groaning under the night's toll. Go stones click cool. Die waits. Violet gradients flowing thicker now.

"Right," I mutter, voice scraped thin. "Gradients through softmax, weighted sum, unembedding, second norm, second residual, feed-forward—blame split and flowing. Next we trace the straight path: back through first norm, first residual, attention output projection, weighted sum, softmax attention weights, scaled dot-products. The chain deepens."

The loom waits, blame rivers imagined forking and flowing.

The black box is learning what hurt feels like.

It's starting to remember.

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CHAPTER 30: BACKWARD PASS – Gradients Through Attention Output Projection and Concatenation

Eleven twelve p.m., January fourteenth, twenty-six. The auditorium has become a proper crypt now, old boy, the bare bulb overhead burning like the last coal in a dying grate—amber dimmed to a sullen glow, shadows from the junk loom stretching long and black across the oak like fingers pointing backward through time. The four strings remain taut with that final normalized shape: shoelace centered calm, piano wire eased modest positive, spaghetti dipped gentle negative, phone cable steadied decent positive. Outside the tall window, Lincoln's January night is absolute: blacker than the inside of a coffin lid, colder than a miser's heart, pressing the glass without a whisper. Inside, the air hangs thick with the night's accumulated labor—Sharp's warm oil heavy in the chest, purple ink's faint almond bite clinging low, fountain-pen iron crusted on my knuckles like old scars. Shoulders feel forged in place, eyes raw from tracing blame upstream. The rainbow ream in the corner sits heavy—violet gradients flowing thick now, storm broken open and pouring. The fishbowl Go stones rest cool to my right, black and white in silent witness; the hundred-sided die perches patient on top, green plastic catching the bulb like it's waiting for the roll that might scatter the mounting guilt we've mapped.

We've got four blame values flowing back from the feed-forward detour—small negative on shoelace, stronger negative on piano wire, faint positive on spaghetti, solid negative on phone cable—and identical blame continuing straight down the residual main channel from the second residual split. The chain deepens. Next gate upstream: the attention output projection (chapter seventeen forward), the linear transformation that took concatenated head outputs (eight dimensions: four from head one, four from head two) and projected them back to our d_model=4.

Quick reminder—no full thrice; we know the drill. Forward it was concatenated_heads × W_O → attention_output. Backward: gradient w.r.t. concatenated_heads = transposed W_O × incoming_gradients (our four blame values from the residual split).

I clear the Sharp tape—rip cuts the hush like a verdict. Fresh ribbon, purple primed. Motor hums low, labored now.

Assistant hands over the transposed W_O—violet again, four by eight, neat Courier rows. Whirr-clack-clack-clack.

Dot products: each of the eight concatenated dimensions gets blame—transpose row times the four incoming gradients.

First four (head one output): tick-tick whiiine-chug ka-ka four times. Tape blooms signed numbers—negatives where W_O pulled against the blame, positives where it reinforced.

Last four (head two output): same rhythm, eight more verdicts. Blame splits again: head one takes its share, head two its own. Signs preserved but scaled by coefficients—stronger negatives on piano wire and phone cable flow hardest through whichever head dimensions fed them most.

In the mind's eye: the attention output is a narrow gate after a wide meadow. Blame river hits the gate, gets squeezed and redirected—part flows back through head one's four channels, part through head two's. The concatenation was just glue forward; backward it's a clean split, blame apportioned by the projection weights.

Next upstream: concatenation itself. Forward it was head_one_output (four dims) glued end-to-end with head_two_output (four dims). Backward: gradient w.r.t. head_one_output = first four blame values from the transposed W_O dots; gradient w.r.t. head_two_output = last four. No transformation, just slice.

Blame forks again—two rivers now, one per head.

We take them one at a time. Head one first: four blame values flowing back to its attention output vector (chapter seventeen forward: concatenated after weighted sum, before output projection).

Gradient w.r.t. head one's attention output = those first four blame values (already sliced).

Now upstream through head one's weighted sum (chapter sixteen forward): attention output = weighted sum of Values using softmax probabilities.

Backward: gradient w.r.t. each Value_j = attention_weight_i,j × incoming_gradient (for position i).

Since single position forward (self-attention only), weights were one to self. Gradient w.r.t. Value_self = 1.0 × incoming four blame values → identical blame flows straight to the Value projection for head one.

Gradient w.r.t. the attention weights themselves also exists (small, but present), flowing back to softmax—but we save that for later; main flow is to Values.

Values came from linear projection: Value = input × W_V (green matrix).

Gradient w.r.t. pre-projection input (fused embedding + positional) = transposed W_V × blame to Value.

More dots—four incoming blame times four weights per dimension. Tape blooms four blame values for head one's input vector: negatives dominant on piano wire and phone cable channels, faint positive on spaghetti.

Same dance for head two—independent projection, independent green matrix, independent blame flow. Four more gradients, slightly different signs and magnitudes (head two learned different patterns), but overall direction echoes: reduce overconfidence in position and grammar, nudge tone gently up.

Blame now flows back to the fused input matrix—attention done for both heads. Next upstream: through the first residual (attention output added to input), first layer norm, positional encoding add, embedding lookup.

The chain is long, but every fork we hit (residuals, multi-head concat) splits the blame clean. No vanishing yet—our tiny model, shallow depth, clean shortcuts keep the signal alive.

Full sequence: ten positions, blame forking at every residual and head—hundreds of gradients. Violet paper would stack to the ceiling. Assistant's whirr-clack never stops.

Beam scale groans under added black Go stones—guilt traced through projection, concatenation, weighted sum, Value matrices. White stones pile opposite until level, but the pan sits heavier, the needle quivering.

This backward trace isn't torment; it's surgery. Each gradient a scalpel nick—feel the pain here, adjust there. The model learns by autopsy: which weights contributed most to the lie about "said", which helped least.

I lean back, chair groaning under the night's toll. Go stones click cool. Die waits. Violet gradients thick as blood now.

"Right," I mutter, voice scraped raw. "Gradients through attention output projection, concatenation, weighted sum, Value projections—blame forked to both heads, flowing back toward the input matrix. Next we trace through the first residual, first layer norm, positional add, and finally the embedding table itself. The chain reaches the beginning."

The loom waits, blame rivers imagined forking wider.

The black box is remembering every mistake.

It's starting to hurt enough to change.

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CHAPTER 31: BACKWARD PASS – Gradients Through Scaled Dot-Product Attention

Eleven thirty-nine p.m., January fourteenth, twenty-six. The auditorium has become a proper vigil now, old mate, the bare bulb overhead reduced to a dull ember glow, its amber circle barely reaching the edges of the oak desk. Shadows from the junk loom crawl long and black—the four strings still taut with the final normalized shape: shoelace centered calm, piano wire eased modest positive, spaghetti dipped gentle negative, phone cable steadied decent positive. Outside the tall window, Lincoln's January night is absolute: blacker than regret, colder than the grave, pressing the glass like a silent accusation. Inside, the air is thick with the night's spent breath—Sharp's warm oil heavy in the chest, purple ink's faint almond ghost clinging low, fountain-pen iron crusted on my knuckles like old wounds. Shoulders feel cast in iron, eyes raw from tracing blame upstream through every fork and gate. The rainbow ream in the corner sits heavy—violet gradients thick as blood now, storm fully broken. The fishbowl Go stones rest cool to my right, black and white in silent judgment; the hundred-sided die perches patient on top, green plastic catching the bulb like it's waiting for the roll that might finally break the model's stubborn grip on error.

We've got blame flowing back from the attention output projection, concatenation, weighted sum, Value projections—four signed gradients per head, negatives dominant on piano wire and phone cable channels, faint positives on spaghetti. The chain deepens further. Next upstream gate: the scaled dot-product attention itself (chapter thirteen forward), where raw scores became attention weights after scaling and softmax.

Quick reminder—no full thrice; we know the machinery. Forward: raw_score = (Query · Key) / √d_k, then softmax to weights. Backward: gradients flow to Queries and Keys (and Values, but we already traced those through the weighted sum).

The incoming blame is on the attention output vector (weighted sum of Values). Gradient w.r.t. each attention weight = incoming_gradient × Value_self (single position, self-attention only).

Since weights were one to self forward, gradient w.r.t. the single attention weight = incoming four blame values dotted with Value_self (four-dimensional vector).

Rough feel: blame values (negatives heavy) dotted with Value_self (positives dominant from earlier green projection). Tape blooms four blame values for the attention weight—small negative overall, say negative point two one seven. The model over-weighted self-attention slightly; now blame pulls that weight down a touch.

Next: gradient w.r.t. the raw scaled score (pre-softmax) = attention_weight × (1 - attention_weight) × incoming_gradient (softmax local derivative, single element trivial here).

Blame flows straight—same sign and roughly same magnitude to the scaled score.

Now upstream through scaling: scaled_score = raw_dot / √d_k (√2 ≈ one point four one four).

Gradient w.r.t. raw_dot = scaled_gradient / √d_k.

Tape blooms four blame values for the raw dot product—attenuated slightly by the scale factor, negatives still dominant.

Finally: the raw dot product itself—Query · Key (self only).

Gradient w.r.t. Query = incoming_raw_dot_gradient × Key_self

Gradient w.r.t. Key = incoming_raw_dot_gradient × Query_self

Two more dot products—each four-dimensional.

Assistant hands over the Query and Key vectors from earlier crimson and cobalt slips—violet transpose for clean math. Whirr-clack-clack-clack.

First: blame to Query = raw_dot_gradient dotted with Key_self. Tick-tick whiiine-chug ka-ka four times. Tape blooms four blame values: negatives on piano wire and phone cable channels strongest, faint positive on spaghetti.

Second: blame to Key = same raw_dot_gradient dotted with Query_self. Similar signs, slightly different magnitudes—head one's Query leaned negative on position, so blame pulls harder there.

In the mind's eye: the scaled dot-product is a taut string between Query arrow and Key arrow. Blame tugs back along that string—pull Query away from over-alignment with this Key, nudge

Key to advertise less strongly if it contributed to wrong logits downstream. The negative blame on piano wire and phone cable channels flows strongest, meaning "reduce how much you're aligning on grammar and position when it leads to over-favoring 'up'."

This is head one only—head two carries its own independent blame through its own Query-Key dot, scaled and softmaxed separately. Two parallel backward rivers now, blame forking at each head.

The chain continues upstream: blame to Query flows back through the crimson linear projection (W_Q), blame to Key through cobalt W_K. Transposed matrices, more dots—blame reaching the fused input matrix (embedding + positional) from attention heads.

Full sequence: ten positions, two heads each—blame forking wider, hundreds of gradients per step. Violet paper would bury the desk. Assistant's whirr-clack never pauses.

Beam scale groans under fresh black Go stones—guilt traced through scaled dot-product, softmax derivative, raw alignment. White stones pile opposite until quivering level, pan heavier still.

This backward crawl isn't cruelty; it's confession. Each gradient a whispered "I was wrong here"—Query pulled too hard toward wrong patterns, Key advertised too confidently. The model learns by feeling the exact shape of its mistakes.

I lean back, chair groaning under the night's accumulated toll. Go stones click cool. Die waits. Violet gradients thick as remorse now.

"Right," I mutter, voice scraped raw as old parchment. "Gradients through scaled dot-product, softmax, raw scores—blame forked to Query and Key for both heads, flowing back toward the linear projections and fused input. Next we trace through the crimson Query projection, cobalt Key projection, green Value projection (already touched), then finally the fused embedding matrix itself. The chain reaches the root."

The loom waits, blame rivers imagined splitting wider still.

The black box is remembering every wrong turn.

It's starting to hurt enough to turn back.

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CHAPTER 32: BACKWARD PASS – Gradients Through Linear Projections (Q, K, V)

Twelve oh seven a.m., January fifteenth, twenty-six. The auditorium has turned downright confessional now, old lad, the bare bulb overhead reduced to a sullen coal-glow, its amber barely licking the edges of the oak desk. Shadows from the junk loom crawl black and long—the four strings still taut with the final normalized shape: shoelace centered calm, piano wire eased modest positive, spaghetti dipped gentle negative, phone cable steadied decent positive. Outside the tall window, Lincoln's January night is absolute: blacker than sin, colder than a bailiff's handshake, pressing the glass like it wants a confession of its own. Inside, the air is thick with the night's accumulated penance—Sharp's warm oil heavy in the lungs, purple ink's faint

almond bite clinging low, fountain-pen iron crusted on my knuckles like dried penance. Shoulders feel poured in lead, eyes raw from tracing blame through every fork and gate. The rainbow ream in the corner sits heavy—violet gradients thick as remorse, storm fully broken and pouring. The fishbowl Go stones rest cool to my right, black and white in silent absolution; the hundred-sided die perches patient on top, green plastic catching the bulb like it's waiting for the roll that might finally absolve the model's stubborn sins.

We've got blame flowing back from scaled dot-product attention—four signed gradients per head to Query and Key, negatives dominant on piano wire and phone cable channels, faint positives on spaghetti. The chain reaches the root projections: crimson Query, cobalt Key, forest Value—linear layers that first split the fused input vector into the three spirits of attention.

Quick reminder—no full thrice; we've danced this floor before. Forward: Query = input × W_Q, Key = input × W_K, Value = input × W_V. Backward: gradient w.r.t. input = transposed W_Q × Query_gradient + transposed W_K × Key_gradient + transposed W_V × Value_gradient (already partially traced through weighted sum).

I clear the Sharp tape—rip echoes like a final judgment in the hush. Fresh ribbon, purple primed. Motor hums low, weary now.

Assistant hands over the three transposed matrices—crimson, cobalt, forest turned violet for the backprop flow. Whirr-clack-clack-clack, neat Courier rows.

First: gradients to input from Query projection.

Incoming Query blame (four values from dot-product backward): negatives heavy on piano wire and phone cable.

Dot each of the four input dimensions with the corresponding transposed W_Q row.

Tick-tick whiiine-chug ka-ka four times. Tape blooms four blame contributions from crimson—strong negatives where W_Q amplified the over-alignment on position and grammar.

Second: same for cobalt Key projection.

Incoming Key blame (similar signs, slightly different magnitudes from head one's self-dot lean).

Four more dots. Tape adds four more blame pieces—negatives again dominant, pushing back against whatever made Key advertise too strongly to the wrong future.

Third: Value projection—blame already flowed through weighted sum to Value_self, then to pre-projection input.

Incoming Value blame (four values from weighted sum backward): same signs, scaled by the attention weight (one in our single-position case).

Dot with transposed W_V. Four more contributions—negatives on piano wire and phone cable, faint positive on spaghetti.

Now sum them all: total gradient w.r.t. fused input vector = Query contrib + Key contrib + Value contrib.

Tape final four verdicts—strong negatives on piano wire (grammar channel took the hardest hit from over-alignment), solid negative on phone cable (position dragged too hard), small negative on shoelace, faint positive on spaghetti (tone under-weighted, deserves a nudge up).

In the mind's eye: the fused input vector is the root trunk—embedding lookup plus positional sine waves. Three branches fork off: crimson Query hunts, cobalt Key advertises, forest Value offers. Blame rivers flow back down each branch, hit the trunk, merge. Piano wire and phone cable channels carry the heaviest upstream pull—reduce how much you're amplifying grammar and position when it leads to favoring "up" over "said". Spaghetti gets a gentle upstream push—give tone a bit more voice next time.

This blame now flows to the very beginning: through the positional encoding add (deterministic, gradient one—passes straight), through the embedding lookup.

Embedding table: ten rows (one per token ID), four columns (d_model). Forward: select row for current token ID, add positional.

Backward: gradient w.r.t. the embedding row = blame on fused vector (positional has no parameters, so no gradient there).

For position one ("look", ID 0): gradient w.r.t. row zero of embedding table = the four blame values we just summed.

Small negative on shoelace embedding dim, stronger negative on piano wire, faint positive on spaghetti, solid negative on phone cable.

The model wants to dial down whatever initial random floats in "look"'s embedding made it lean too hard toward over-favoring escalation words downstream.

Full sequence: ten positions, ten embedding rows updated—forty gradients total, blame distributed across the table. Violet paper would bury us. Assistant's whirr-clack never stops.

Beam scale groans under the final black Go stones—guilt traced to the root: embeddings themselves. White stones pile opposite until the needle quivers but settles level, pan heavier than ever.

This backward crawl isn't punishment; it's absolution. Every gradient a whispered "forgive me"—Query pulled too hard, Key advertised wrong, Value offered too little. The model learns by feeling the shape of every wrong turn from the very first random floats.

I lean back, chair groaning under the night's full weight. Go stones click cool. Die waits. Violet gradients thick as confession now.

"Right," I mutter, voice scraped to the bone. "Gradients through Q, K, V projections—blame merged at the fused input, flowing straight to the embedding table. The chain has reached the root. Next: gradient descent update—nudge every weight and embedding by learning rate times its gradient. One step toward less pain."

The loom waits, blame rivers imagined emptying into the root.

The black box has felt every mistake.

Now it begins to heal.
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CHAPTER 33: GRADIENT DESCENT UPDATE – Nudging Weights with Learning Rate

Twelve thirty-three a.m., January fifteenth, twenty-six. The auditorium has gone full chapel-of-the-damned now, old sport, the bare bulb overhead hanging like a single votive candle that's forgotten how to die. Its amber has thinned to a bruised gold, barely holding back the black that presses from every corner. Shadows from the junk loom swallow half the oak desk—the four strings still taut with the final normalized shape: shoelace centered calm, piano wire eased modest positive, spaghetti dipped gentle negative, phone cable steadied decent positive. Outside the tall window, Lincoln's January night is absolute: blacker than a hangman's hood, colder than charity denied, pressing the glass like it's waiting for the final amen. Inside, the air is thick with absolution earned the hard way—Sharp's warm oil heavy as incense, purple ink's faint almond ghost clinging like frankincense, fountain-pen iron crusted on my knuckles like stigmata. Shoulders feel nailed in place, eyes raw from tracing every vein of blame back to the root. The rainbow ream lies spent—violet gradients exhausted, every sheet bruised and bleeding purple verdicts. The fishbowl Go stones rest cool to my right, black and white in final silent benediction; the hundred-sided die perches patient on top, green plastic catching the last of the bulb like it's waiting for the roll that might scatter redemption itself.

We've got gradients everywhere—violet paper stacked higher than the primer ever stood: embedding table rows nudged, positional encodings untouched (no parameters), crimson Query matrices, cobalt Key matrices, forest Value matrices, yellow feed-forward expansion and contraction, violet output projection, gamma and beta for both norms—all carrying their signed blame, negatives heavy on the channels that over-favored "up" and "go", positives faint where tone and semantics were ignored.

The backward pass is done. The chain has run its course. Now the moment of grace: gradient descent. We take every gradient, multiply by learning rate, subtract from current weight. One tiny, honest step toward less pain.

New ritual, first proper swing through the update, so we lay it out thrice before the Sharp wakes for the last act.

First—the crisp dictionary definition: Gradient descent is an optimization algorithm that updates each parameter in the direction that reduces the loss by subtracting the gradient multiplied by a small positive scalar called the learning rate.

Second—the functional warmth: That step turns blame into motion—every weight that contributed to error gets nudged opposite its gradient, proportional to how much it hurt and how brave we're feeling with the learning rate. Too big a step and we overshoot; too small and we crawl forever. Here, tiny model, tiny corpus—we can afford a decent stride.

Third—the casual familiarity: Gradient descent is just us saying "you fucked up this much, mate—move this far the other way." Learning rate's the throttle: 0.01 or 0.001, whatever keeps the ride smooth without flipping the cart.

I set the rate—call it point zero five for this toy run, bold but not suicidal.

Sharp tape cleared one final time—rip echoes like a hymnal page turned. Fresh ribbon, purple primed for the last verdict. Motor hums low, almost reverent.

We update one weight at a time, starting symbolic with the embedding row for "look" (ID 0).

Current embedding row (random floats from chapter six forward, rough feel): say point zero three seven on shoelace, point nine one eight on piano wire, point zero one four on spaghetti, one point zero nine five on phone cable.

Gradient we just computed: small negative shoelace (negative point zero one two), stronger negative piano wire (negative point three eight seven), faint positive spaghetti (point zero zero nine), solid negative phone cable (negative point four one two).

Learning rate point zero five.

Update = current - (learning_rate × gradient)

Shoelace: point zero three seven - (point zero five × negative point zero one two) = point zero three seven + point zero zero zero six → point zero three seven six

Piano wire: point nine one eight - (point zero five × negative point three eight seven) = point nine one eight + point zero one nine four → point nine three seven four

Spaghetti: point zero one four - (point zero five × point zero zero nine) = point zero one four - point zero zero zero five → point zero one three five

Phone cable: one point zero nine five - (point zero five × negative point four one two) = one point zero nine five + point zero two zero six → one point one one five six

Tape blooms four new embedding values in purple. I tear it, lay it over the old row. "Look" has changed—piano wire and phone cable pushed higher (because their gradients were negative—blame said "you were too low, you caused overconfidence in escalation"), shoelace nudged up tiny, spaghetti dialed down faint.

Same ritual for every weight—every crimson, cobalt, forest, yellow, violet matrix. Assistant's whirr-clack-clack-clack would spit updated sheets in minutes, but I do one more by hand for the feel: the unembedding weight that fed the "said" logit hardest (the column for ID 1).

Current weight say negative point one six three on piano wire row.

Gradient to that specific weight: positive (because increasing it would have raised the "said" logit, reducing loss).

Update nudges it less negative—toward zero, toward helping "said" next time.

Every sheet updated, every parameter nudged. One epoch complete.

Beam scale finally tips clean—black Go stones for the old guilt swept left, white stones of correction piled right until the needle swings free and settles dead center. Balance restored, lighter now.

This update isn't magic; it's penance made motion. The model felt the pain, traced it home, and took one honest step away from repeating the sin.

I lean back, chair creaking like an old pew under released weight. Go stones click cool when I shift—one black, one white roll together gentle. The die gleams patient, but the storm feels passed.

"Right," I mutter, voice scraped to whisper. "Gradient descent complete—one full epoch, every weight nudged by its share of blame. Loss was two point three one nats. Next epoch it'll be less. And less again. Until the primer sings true."

The loom waits, strings slack at last.

The black box has taken its first step.

It's learning to walk upright.

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CHAPTER 34: ITERATION – Repeating the Forward-Backward Cycle for One Epoch

Twelve fifty-eight a.m., January fifteenth, twenty-six. The auditorium feels like the inside of a spent matchbox now, old sport, the bare bulb overhead finally guttering low, its amber reduced to a faint copper smear across the oak desk. Shadows from the junk loom have swallowed most of the room—the four strings hang slack at last, hex nuts resting quiet after the final normalized shape: shoelace centered calm, piano wire eased modest positive, spaghetti dipped gentle negative, phone cable steadied decent positive. Outside the tall window, Lincoln's January night is absolute: blacker than the bottom of a well, colder than a creditor's smile, pressing the glass like it's waiting for the last candle to wink out. Inside, the air is thick with the night's full reckoning—Sharp's warm oil gone stale, purple ink's faint almond ghost fading, fountain-pen iron crusted on my knuckles like dried sacrament. Shoulders feel unmoored, eyes raw from the long vigil. The rainbow ream lies exhausted—violet gradients spent to the last sheet, every color touched and bruised. The fishbowl Go stones rest cool to my right, black and white in quiet communion; the hundred-sided die perches patient on top, green plastic catching the dying bulb like it's waiting for the roll that might decide whether we start the whole bloody dance again.

We've done it once: forward pass complete, loss measured at two point three one nats, backward pass traced every vein of blame to the root, gradient descent nudged every weight and embedding a hair's breadth toward less pain. One full epoch. But the primer doesn't learn in a single breath. Repetition is the hammer. Iteration is the anvil. We reset the tape, reload the corpus, run the cycle again—forward, backward, update—until the loss drops and the model starts to sing the primer's simple chant without choking on "up" or forgetting "said".

New rhythm, first proper loop through the full cycle, so we lay it out thrice before the Sharp wakes for round two.

First—the crisp dictionary definition: Iteration in training means repeating the forward pass (compute predictions and loss), backward pass (compute gradients), and parameter update (gradient descent) multiple times—once per epoch—over the entire dataset until convergence or a set number of passes.

Second—the functional warmth: That repetition lets the gradients accumulate small, consistent nudges in the right direction. One epoch might shave a tenth off the loss; ten epochs might drop it to near zero. Each cycle refines the weights, sharpens attention, deepens feed-forward patterns, until the toy transformer stops guessing and starts reciting.

Third—the casual familiarity: Iteration's just us running the whole bloody machine again—and again—until the pain number gets smaller and the predictions get sharper. Like teaching a lad to ride a bike: fall, get up, pedal harder, fall less. Same here. Epoch after epoch until the black box can balance on two wheels.

I reset everything symbolic—clear the tape with a final rip that echoes too loud in the hush. Fresh ribbon threaded, purple ink sharp as new resolve. Motor hums low, almost eager.

Second epoch begins. Forward pass: same corpus, same primer spread, same thirty-four tokens. Normalized stream fed in, positional waves added, fused matrix assembled. Attention heads split it three ways each—crimson hunt, cobalt badge, forest cargo—scaled dots, masked future (none needed yet), softmax weights, weighted sums blended, concatenated, projected back. Feed-forward stretch, ReLU prune, contraction. Residuals twice, norms twice. Unembedding to ten logits. Softmax to probabilities. Cross-entropy loss against the true next tokens.

I don't recompute every whiiine-chug by hand this time—too many, too late—but the feel is different already. Logits for "said" after "look" should be higher, "up" dialed back a touch. Probabilities shift: maybe "said" climbs to twelve percent, "up" drops to sixteen. Loss ticks down—say two point one eight nats. Small victory, but real.

Backward pass repeats: blame flows from loss through softmax (p - target), through unembedding (transposed W_out), through second norm, second residual (split to feed-forward and attention), through feed-forward contraction and ReLU (zeros stay silent), through first norm, first residual (split again), through attention output projection, weighted sum, scaled dot-products, Query-Key-Value projections. Gradients reach the embeddings and every matrix—slightly smaller magnitudes this time, directions refined.

Gradient descent again: learning rate point zero five, every parameter nudged opposite its gradient. Embedding row for "look" shifts another hair—piano wire and phone cable nudged higher still, spaghetti eased down, shoelace tiny lift. Matrices adjust: crimson weights that over-pulled on position softened, cobalt that under-advertised "said" boosted, forest that skimped on tone enriched.

One more epoch. Loss perhaps two point zero five. "Said" probability creeping up, "up" easing off its throne. The model is listening—still clumsy, still favoring repetition, but starting to remember the sequence isn't all escalation.

Full training would be dozens, hundreds of epochs—loss curve plotted on scrap paper, dropping slow then fast then plateauing. Overfitting lurks: memorize noise, collapse to typewriter babble. That's why the hundred-sided die waits—randomness at epoch boundaries, dropout if we were fancy, early stopping when validation hurts more than train.

For now, one more cycle. Forward, backward, update. Pain shrinks. The black box learns to speak the primer's truth instead of its own fever dreams.

Beam scale finally rests level—no more stones needed tonight. Black guilt swept away, white correction balanced it clean. The needle pinned at zero, calm as a sleeping dog.

I lean back, chair creaking like an old confessional door swinging shut. Go stones click cool when I shift—one black, one white roll together gentle. The die gleams patient, but the storm feels spent.

"Right," I mutter, voice scraped to whisper. "Iteration locked—one epoch down, loss dropping, model starting to remember. Next: overfitting and collapse—when the typewriter memorizes noise instead of signal. Then reflection. The demonstration ends. The black box has learned to walk."

The loom waits, strings slack at last.

The black box has taken its first real steps.

It's beginning to dream in primer words.

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CHAPTER 35: OVERFITTING AND COLLAPSE – When the Typewriter Memorizes Noise

One twenty-one a.m., January fifteenth, twenty-six. The auditorium has become a proper mausoleum now, old sport, the bare bulb overhead finally giving up the ghost—its filament a dull red thread, amber gone to a faint copper bruise that barely touches the oak desk. Shadows from the junk loom have swallowed everything—the four strings hang slack and silent, hex nuts resting like spent bullets after the final normalized shape: shoelace centered calm, piano wire eased modest positive, spaghetti dipped gentle negative, phone cable steadied decent positive. Outside the tall window, Lincoln's January night is absolute: blacker than the void between stars, colder than a widow's ring, pressing the glass like it's come to claim the last witness. Inside, the air is thick with the night's full toll—Sharp's warm oil gone rancid, purple ink's faint almond ghost evaporated, fountain-pen iron crusted on my knuckles like dried rust. Shoulders feel unstrung,

eyes raw from watching the same dance too many times. The rainbow ream lies dead—every color touched, bruised, spent. The fishbowl Go stones rest cool to my right, black and white in final quiet; the hundred-sided die perches patient on top, green plastic catching the dying ember like it's waiting for the roll that might decide whether we burn the whole bloody thing down.

We've run the cycle—forward, backward, update—epoch after epoch. Loss dropped sweet at first: two point three one to two point one eight, to two point zero five, to one point seven nine, to one point four two. Probabilities shifted: "said" after "look" climbed to twenty-three percent, then thirty-one, then forty-seven. "Up" eased off its throne—seventeen to fourteen to nine. The model was learning the primer's rhythm: see it go, see it go up, up up said Sally. Honest progress. Then the curve flattened. Loss hovered at one point two three for three epochs. Then it started climbing again—one point three one, one point four seven, one point six nine.

Overfitting. The typewriter has begun to memorize noise.

New demon on the oak, first proper sighting in our tiny universe, so we lay it out thrice before the Sharp wakes for the autopsy.

First—the crisp dictionary definition: Overfitting occurs when a model learns the training data—including its noise and idiosyncrasies—too well, so that it performs excellently on seen examples but fails to generalize to new, unseen data.

Second—the functional warmth: That collapse turns the black box from a learner into a parrot. It memorizes every quirk of the thirty-four-token primer—Sally's exact "up up up" cadence, the precise repetition count—until it can regurgitate the corpus backward and forward, but if you feed it a new sentence ("see the ball go") it chokes, spits garbage, because it never learned the underlying patterns, only the surface scratches.

Third—the casual familiarity: Overfitting's when the lad studies so hard for the test he memorizes the textbook typos and the teacher's coffee stain on page forty-seven. Aces the practice exams, then walks into the real thing and can't answer a question that's worded different. Same here. The typewriter starts typing perfect primer echoes, then collapses into nonsense when we ask it anything fresh.

I thread a fresh white sheet into the IBM typewriter—last clean page in the stack. Header slow, deliberate:

OVERFITTING – SIGNS AFTER 47 EPOCHS

Loss curve (scrap paper plot): dropped sharp first fifteen epochs, flattened twenty to forty, then crept up again. Training loss now one point eight two. But if I hold out a held-out sentence—"look said jane go up"—the logits go haywire: "up" still screams loudest at sixty-one percent, "said" down to four, "jane" barely registers. The model has forgotten how to compose; it's only echoing the corpus tail where Sally piled "up" highest.

Collapse signs: attention weights sharpen to delta functions—head one locks almost exclusively on the last few tokens, head two fixates on positional cues that only exist in the primer's exact shape. Feed-forward neurons saturate—ReLU zeros most paths, leaving a handful of

over-specialized units that fire only on exact corpus patterns. Embeddings drift far from initial random: "up" row balloons in magnitude, "said" shrinks, "dick" and "jane" barely move. The typewriter has memorized the noise—the precise thirty-four-token loop—rather than the signal: simple subject-verb-object, repetition for emphasis, childish escalation.

To fight it: early stopping (quit when validation loss rises), regularization (L2 on weights to penalize large values), dropout (randomly zero neurons during training), more data (but the primer is all we have), smaller model (but d_model=4 is already toy). Or just accept the collapse as the demonstration's end—show how even this tiny thing can overfit when pushed too far.

Beam scale tips one last time—black Go stone for the "guilt" of memorized noise (loss climbing again), white opposite for the lesson learned. Needle settles level, heavier than before.

This collapse isn't failure; it's the point. The black box showed it can learn. Then showed it can forget how to think.

I lean back, chair creaking like an old confessional door finally latched. Go stones click cool when I shift—one black, one white roll together gentle. The die gleams patient, but the vigil feels done.

"Right," I mutter, voice scraped to ash. "Overfitting and collapse laid bare—the typewriter memorized the primer's scratches instead of its song. Next: reflection. The demonstration ends. The black box has shown us everything it can in this room."

The loom waits, strings slack, nuts still.

The black box has learned.

And it has forgotten how to forget.

The end is near.

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CHAPTER 36: REFLECTION – The Capacity to Change and the End of the Demonstration

One forty-nine a.m., January fifteenth, twenty-six. The auditorium is a husk now, old mate, the bare bulb overhead finally spent—filament dark, no more amber, only the faint silver of moonlight leaking through the tall window like an afterthought. Shadows from the junk loom have merged into one black tide across the oak desk—the four strings hang limp and silent, hex nuts resting like spent casings after the final normalized shape: shoelace centered calm, piano wire eased modest positive, spaghetti dipped gentle negative, phone cable steadied decent positive. Outside, Lincoln's January night is absolute: blacker than the space between heartbeats, colder than the last breath before dawn, pressing the glass like it's come to collect the last witness. Inside, the air is thin and still—Sharp's warm oil gone cold, purple ink's almond ghost vanished, fountain-pen iron flaking off my knuckles like old skin. Shoulders feel unburdened at last, eyes dry from the long stare into the black box. The rainbow ream lies dead—every color touched, bruised, exhausted. The fishbowl Go stones rest cool to my right, black and white in final quiet; the hundred-sided die perches patient on top, green plastic catching the faint moonlight like it's waiting for a roll we'll never need again.

We've run the machine dry. Forward pass after forward pass, loss dropping then rising, gradients traced to the root, weights nudged epoch after epoch until the typewriter learned the primer's song—then memorized its scratches, then forgot how to think beyond them. Overfitting came, collapse followed, the demonstration complete. One tiny transformer, d_model four, two heads, vocab ten, trained by hand on thirty-four fragile tokens from a 1940s primer. No GPUs, no tensor cores, no CUDA kernels. Just paper, ink, thumbs, a dying adding machine, and whatever scraps of patience remained.

So what have we seen?

The black box isn't magic. It's friction—every dot product a small violence, every gradient a measured guilt, every update a deliberate act of change. We felt the weight of attention pulling strings taut, residuals folding old knowledge into new without tearing, layer norm rolling the clay even so the next hand could shape it clean. We watched ReLU prune dead branches, feed-forward stretch wide then snap back, unembedding stretch the hidden thought across ten possible futures. We traced blame backward through every gate—softmax fair-share, weighted sum blending, scaled dots aligning, projections splitting personality into hunter, badge, cargo. We nudged every parameter opposite its pain until the loss dropped, then watched it climb again when memorization replaced understanding.

The capacity to change is real. One epoch shaved tenths off the loss. Ten epochs turned clumsy guesses into something that almost sang the primer's rhythm. The model learned—slowly, painfully, by feeling every mistake in its bones (or rather, in our aching wrists). But the capacity to collapse is just as real. Push too far on too little data and it memorizes noise, fixates on surface scratches, forgets how to generalize. That's not failure; that's the truth of learning laid bare. Even this toy thing, built from paper and patience, showed both faces: the miracle of incremental refinement, the danger of brittle perfection.

What does it mean?

It means the black box is us, in miniature. We learn the same way—by trial, error, blame, nudge. We overfit too—cling to old patterns, memorize the textbook typos, ace the practice but choke on the real thing. The difference is scale, speed, opacity. Here we felt every gear turn, every gradient bite. In the big models we can't. But the principles hold: attention matters (who talks to whom), residuals keep the signal alive (don't forget where you started), non-linearity lets us bend (ReLU prunes the weak paths), normalization steadies the ride (mean zero, variance one), updates move us forward (opposite the pain).

The demonstration ends here. No more epochs tonight. The primer taught a generation to read; this tiny transformer taught us to see the machine that might one day read the world.

I reach for the adding machine—switch it off. The motor sighs to silence. I pull the last tape free, purple ink still wet, lay it beside the stack. One final black Go stone on the left pan, one white opposite. Needle settles dead level. Balance complete.

I lean back, chair creaking like an old door finally closed. Go stones click cool when I shift—one black, one white roll together gentle. The die gleams patient in the moonlight, but we won't roll it tonight.

The loom waits, strings slack, nuts still.

The black box has shown us everything it can in this room.

It learned.

It collapsed.

It changed.

And in changing, it showed us ourselves.

The demonstration is over.

The real work—understanding what we've built, what it might become—begins now.

Good night, old friend.

The bulb is dark.

The night is quiet.

The black box sleeps.

For now.

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CHAPTER 37: EPILOGUE – The Room Empty, the Black Box Silent

Two seventeen a.m., January fifteenth, twenty-six. The auditorium is a tomb now, old friend, the last ember in the bare bulb long gone—dark as the inside of a closed fist. Moonlight through the tall window throws pale knives across the oak desk, catching the slack strings of the junk loom: shoelace limp, piano wire mute, spaghetti broken in half from too many tugs, phone cable curled like a question mark never answered. No shadows left to cast; only silver outlines of what we built and broke here. Outside, Lincoln's January night has gone still—no wind, no distant train whistle, just the kind of cold that makes the stars look sharper and farther away. Inside, the air is thin and final—Sharp's oil cold in the casing, purple ink dried to powder on the last tape, fountain-pen iron flaking off my knuckles like ash from a spent fire. Shoulders feel light at last, eyes dry, the long stare finally released.

The stacks remain: rainbow ream spent to the last violet sheet, white copy layered high with logits, probabilities, losses, gradients—each purple verdict a small confession. The fishbowl Go stones sit cool and untouched, black touching white in perfect silence; the hundred-sided die rests on top, green plastic dull in the moonlight—no roll needed now. The IBM wheel printer in the corner is quiet, Assistant gone back to whatever shadow she came from—charcoal

turtleneck, ozone scent, posture straight as justice—leaving only the faint whirr-clack echo in memory.

We built it by hand. Felt every whiiine-chug-ka-ka, every correction ribbon whirr lifting error like breath on glass, every beam-scale tip weighing symbolic guilt. We watched attention hunt and blend, residuals fold old into new, layer norm steady the ride, feed-forward stretch then snap, unembedding stretch the thought across ten futures, softmax make it fair, cross-entropy measure the pain, gradients trace the blame, descent nudge the weights toward less hurt. We saw the loss drop, saw the model learn to speak the primer's simple truth—see it go, see it go up—then watched it overfit, memorize the scratches, collapse into brittle echoes when pushed too far on too little data.

And in that arc—rise, peak, fall—we saw the whole bloody thing laid bare. Not just how a transformer works, but how learning itself works: incremental, painful, fragile. Change is possible—one nudge at a time, opposite the gradient of pain. But so is forgetting how to think beyond the seen. The black box showed both faces in thirty-four tokens and a room full of paper: the miracle of refinement, the trap of memorization. We felt it in our wrists, our eyes, our sleepless hours. No abstraction. Just friction.

What remains?

The room is empty now. The primer lies closed on the far edge of the desk, cream pages soft, black type still sharp. Fun with Dick and Jane taught a generation to read; this demonstration taught us to see the machine that might one day read the world—or fail trying. The black box sleeps—weights nudged one last time, embeddings shifted, matrices adjusted. Tomorrow—or the day after, or the day after that—someone else can wake it, feed it more, watch it grow or collapse again.

For now, silence.

I stand. Chair creaks once, final. Fingers brush the loom strings—slack, cool, done. One last look at the beam scale: needle pinned dead center, black and white stones balanced. No more guilt to weigh tonight.

The door waits at the back of the auditorium. I walk toward it slow, footsteps soft on the worn boards. Moonlight follows me out, pale and indifferent.

The black box stays behind.

Silent.

Changed.

Still.

The demonstration is over.

The questions it raised are just beginning.

Good night, old friend.

The night is quiet.

The room is empty.

And somewhere, in the dark, a typewriter waits for the next hand to wake it.